Journal of Cheminformatics

REVIEW Open Access

# Chemical space visualization at scale: a survey of end-to-end pipelines and dataset-size archetypes

Maha M. AlShammari[1,2*], Moayad Alnammi[1,3] and Moataz Ahmed[1,4]

## Abstract

Chemical space visualization supports exploration of high-dimensional molecular data by revealing patterns of similarity, diversity, and structure–property relationships. As chemical libraries expand from thousands to billions of compounds, practical visualization increasingly depends on pipelines that balance chemical meaning with computational and memory constraints. In this survey, we review 56 studies published between 2000 and 2025 and synthesize the end-to-end workflow of chemical space visualization across five core stages: dataset choice, molecular featurization, dimensionality reduction (DR), clustering, and evaluation. We quantify usage trends over time and relate method selection to dataset scale. Across the literature, fingerprints remain the dominant representation for large libraries due to their scalability, while recent studies increasingly incorporate fragments, SMILES-based encodings, and learned embeddings when richer signals are needed. DR practice shows a shift from PCA-centric baselines to neighborhood-preserving methods such as t-SNE and UMAP, with graph layout approaches like TMAP enabling visualization at extreme scale. Clustering shows the weakest convergence to a single standard: hierarchical methods, K-means, and SOM remain frequent choices, complemented by scalable summarization and domain-driven strategies (e.g., BIRCH/BitBIRCH and scaffold-based partitioning) when all-pairs similarity becomes prohibitive. Finally, we propose dataset-size-aware pipeline archetypes and identify open challenges, including inconsistent structure-aware evaluation, limited reproducibility reporting, and the need for scalable, chemically grounded methods for ultra-large libraries.

**Keywords** Chemical space visualization, Molecular representation, Fingerprints, Dimensionality reduction, Clustering, Scalability, Pipeline, Cheminformatics

## Introduction

Chemical space denotes the set of all theoretically possible molecular structures and is vast even under conservative constraints; for example, the number of small organic molecules with up to 30 atoms has been estimated to lie between $10^{60}$ and $10^{100}$ [1]. The scale of this space makes systematic exploration a central challenge in drug discovery and materials design, where the practical goal is to search large collections of molecules, identify promising regions, and prioritize candidates for downstream validation [2]. This need is amplified by the rapid growth of modern chemical libraries that now reach tens of billions

*Correspondence:
Maha M. AlShammari
mmashammari@iau.edu.sa
[1] Information and Computer Science Department, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia
[2] Computational Unit, Department of Environmental Health, Imam Abdulrahman Bin Faisal University, Dammam 31441, Saudi Arabia
[3] Interdisciplinary Research Center for Intelligent Secure Systems, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia
[4] SDAIA-KFUPM Joint Research Center for Artificial Intelligence, King Fahd University of Petroleum and Minerals, Dhahran 31261, Saudi Arabia

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

of compounds (e.g., ZINC22), increasing the demand for scalable computational methods that can organize and interpret high-dimensional molecular representations [3].

Chemical space visualization provides a practical way to support this exploration by constructing low-dimensional maps that reveal structure, similarity, and diversity patterns. Most visualization workflows follow three core steps. First, featurization converts molecular structures into numerical representations, including descriptors, fingerprints, graph-based encodings, Simplified Molecular Input Line Entry System (SMILES)-derived features [4], and learned embeddings [5]. Second, dimensionality reduction (DR) maps these high-dimensional representations into 2D or 3D spaces to enable visual inspection and comparative analysis of molecular neighborhoods [6–8]. Third, clustering groups molecules into chemically meaningful regions of the map, supporting tasks such as library summarization, series identification, and diversity analysis; common choices include K-means, hierarchical clustering, and self-organizing maps (SOM), with scalable alternatives (e.g., BIRCH/BitBIRCH) used when dataset size and compute constraints require it [9]. The quality of these groupings and projections is assessed using complementary evaluation signals, including structural metrics (e.g., Tanimoto similarity and scaffold analysis), statistical validity indices (e.g., Silhouette, Davies-Bouldin Index (DBI), Calinski-Harabasz Index (CHI), Adjusted Rand Index (ARI)), and projection-fidelity metrics that quantify neighborhood preservation in the low-dimensional map [10].

A number of reviews have discussed parts of this workflow, including early surveys of classical projection methods [1, 11], computational mapping tools for drug discovery [12], visual representations of chemical space spanning multiple method families [13], navigation of natural product chemical space [14], open-source visualization and chemical space analysis tools [15, 16], and a recent survey of DR algorithms and software for chemical space navigation [17]. Table 1 summarizes these representative reviews. However, these existing reviews typically focus on specific methods or application domains rather than examining the full visualization pipeline from featurization through DR and clustering. In particular, they rarely connect method selection to dataset scale and computational constraints in a systematic way.

In this work, we address these gaps by providing an integrative review of chemical space visualization workflows spanning datasets, featurization, DR, clustering, and evaluation. We analyze 56 studies published between 2000 and 2025 to (i) quantify usage trends over time, (ii) relate method choices to scalability constraints using computational and memory complexity summaries, and (iii) propose dataset-scale pipeline archetypes that translate literature trends into practical workflow templates. Together, these contributions provide a consolidated reference for selecting and justifying chemical space visualization methods under realistic data and compute regimes.

## Review methodology

Our review methodology follows a structured approach consisting of four main stages: research questions, search strategy design, study selection, and data synthesis, and is guided by the Modular Chemical Space Visualization Pipeline illustrated in Fig. 1. The main goal of this study is to provide a comprehensive review of state-of-the-art approaches for chemical space visualization, with a particular focus on how DR and clustering techniques are integrated within modern computational pipelines.

The illustrated pipeline comprises five consecutive components: dataset selection, molecular featurization,

**Table 1** Representative reviews related to chemical space visualization

| Year | Review | Brief Overview |
|---|---|---|
| 2003 | Daszykowski et al. [11] | Survey of DR methods for chemical data (linear and nonlinear) and their visualization trade-offs |
| 2008 | Medina-Franco et al. [1] | Chemical space visualization in drug discovery and across compound collections |
| 2009 | Ivanenkov et al. [12] | Computational mapping tools for drug discovery, focusing on nonlinear DR and SOM approaches |
| 2012 | Lachance et al. [14] | Charting and navigating natural product chemical space and its implications for drug discovery |
| 2015 | Osolodkin et al. [13] | Visual representations of chemical space, spanning multiple method families (e.g., descriptor plots, Principal Component Analysis (PCA), Self-Organizing Maps (SOM), Generative Topographic Mapping (GTM), and graph approaches) |
| 2022 | Medina-Franco et al. [15] | Progress on open chemoinformatics tools for exploring chemical space, including representations and visualization capabilities |
| 2022 | Saldívar-González et al. [16] | Approaches for chemical space analysis in drug discovery, covering navigation methods, diversity assessment, constellation plots, and open-source visualization tools |
| 2025 | Sosnin, Sergey [17] | Survey of DR algorithms, software tools, and applications for chemical space visualization, with emphasis on scalability, parametric methods, and interactive exploration |

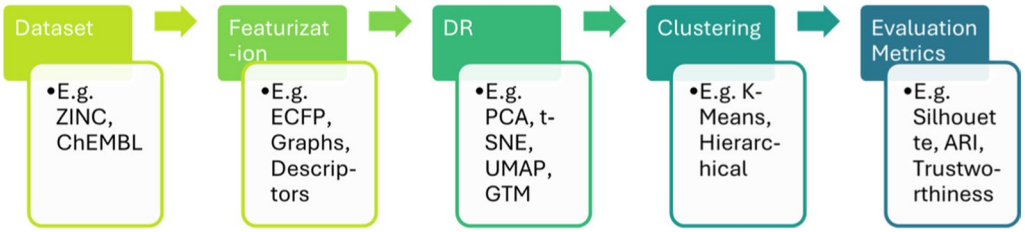



**Fig. 1** Modular pipeline for chemical space visualization: an overview of the five core stages reviewed in this survey

DR, clustering, and evaluation metrics. Although the pipeline is presented as a sequence of conceptually distinct stages, some techniques span multiple stages. In particular, methods such as Self-Organizing Maps (SOM) and Generative Topographic Mapping (GTM) perform DR and implicit clustering simultaneously. In such methods, latent nodes or map units identify regions that can be interpreted as clusters, and these regions can be analyzed in terms of chemical property distributions or aggregated into higher-level superclusters. Consequently, the separation of DR and clustering in this review is organizational rather than methodological. Together, these components provide a basis for methodological comparison across studies. In this review, we focus on addressing the following research questions (RQ):

- **RQ1**: What datasets have been used for chemical space visualization, and what are their scales?
- **RQ2**: Which featurization techniques (e.g., molecular descriptors, fingerprints, and graph-based representations) have been used? What are the recent trends among them during the selected period? And how do their representations impact the downstream tasks?
- **RQ3**: What are the recent trends in DR for chemical space visualization? And what are their limitations?
- **RQ4**: What are the recent trends in clustering algorithms for chemically meaningful groupings? What are their strengths and shortcomings? And what evaluation metrics were used to assess their quality?
- **RQ5**: What scalable computational strategies exist for DR and clustering of large-scale chemical datasets?
- **RQ6**: What gaps exist in the current literature, and what future research directions could help bridge these gaps?

## Search strategy

The search process was conducted in Scopus, Google Scholar, and ACS Publications for studies published between 2000 and the beginning of 2025, using the following search string:

*("Chemical Space" OR "Large Chemical Space" OR "Compounds Database") AND ("Visualization" OR "Dimensionality Reduction" OR "Clustering" OR "Cheminformatics")*

This period was selected due to the rise of public molecular databases (e.g., ChEMBL, PubChem, ZINC) and modern DR methods (e.g., t-SNE and Uniform Manifold Approximation and Projection (UMAP)) that emerged and were employed in the field after 2000. Therefore, the selected period might capture the evolution of computationally driven and large-scale chemical space visualization.

## Study selection

We intentionally applied broad search queries to ensure comprehensive coverage, which retrieved a large number of studies requiring further filtering. To align these studies with our research questions, we identified the most relevant ones via a multi-stage selection process. This process started with screening the title, abstract, and content to ensure compatibility with the goals of the study. Studies were included if they explicitly addressed chemical space visualization or related workflows involving molecular representation, DR, or clustering. Preference was given to studies providing methodological descriptions, dataset information, and quantitative analysis related to chemical space exploration. Studies focused on unrelated applications (e.g., predictive QSAR modeling without visualization, or purely synthetic chemistry reports) were excluded. This filtering process ensured that the final set of selected studies directly contributed to the understanding of method choices and trends in chemical space visualization. These studies emphasize methodological clarity regarding datasets, featurization, and evaluation metrics. To account for both established and new trends, recent works and preprints introducing new approaches were also included. Overall, 56 studies met the inclusion criteria for detailed analysis.

## Data synthesis

Adhering to the five components in the developed pipeline (Fig. 1), the included studies were analyzed along five dimensions:

- *Used Dataset*: includes the dataset name, source, and size range.
- *Featurization*: summarizes the frequently used molecular representation techniques and their impact.
- *DR*: compares the frequently used DR techniques based on their families: linear, nonlinear, and latent variable models.
- *Clustering*: compares the frequently used techniques based on their families: partitional, hierarchical, and soft clustering algorithms, as well as scalable strategies incorporated into them.
- *Evaluation Metrics*: based on their usage frequency, we group them into statistical, structural, and projection-fidelity categories.

Additionally, for the top-trending DR and clustering methods, we have conducted a comparative analysis with an emphasis on their strengths, weaknesses, and computational nature, including time and space complexity, to shed more light on their scalability and applicability to large-scale chemical data. The detailed comparison results for the main attributes of the 56 selected studies are shown in Table 2.

## Employed datasets in chemical space visualization

Chemical space visualization relies on datasets that differ widely in source and scale, which strongly influence the analysis and methods that are feasible. From the reviewed studies, four sources dominate: public repositories that pair structures with assay annotations (e.g., ChEMBL [69], PubChem [70], DrugBank [71]), very large on-demand purchasable libraries used to chart the breadth of accessible chemical space (e.g., ZINC [72] and make-on-demand collections), in-house high-throughput screening collections, and domain-specific collections (e.g., natural-product, metabolomics, and materials datasets). Across these sources, visualizations are used to characterize structural diversity and scaffold coverage, compare libraries, guide library design and prioritization, and benchmark DR and clustering pipelines on datasets ranging from thousands to billions of molecules. This section addresses **RQ1** by summarizing the sources and scales of datasets used in the 56 reviewed studies.

### Sources of datasets

Chemical space analyses commonly draw on well-established public resources such as ChEMBL [30, 35, 44, 46, 47, 51, 52, 56, 57, 66, 67], ZINC [3, 9, 24, 37, 46], PubChem [9, 35], and DrugBank [29, 63]. The dataset sizes reported in this subsection reflect those used in the cited studies at the time of their analysis and may differ from current database releases. ZINC, one of the largest virtual-screening resources, includes billions of purchasable molecules, with ZINC22 providing one billion compounds for large-scale interaction and diversity studies [9]. ChEMBL (1.56 million compounds) is central to structure-activity modeling and target-centric analyses [35]. PubChem hosts millions of molecular structures and annotations for structural and property-based analyses [9], while DrugBank offers curated data on approved and investigational drugs essential for visualizing therapeutic chemical space [29]. Specialized datasets frequently used for benchmarking include SMU (8.9 million compounds) [25], the NCI AIDS dataset (29,184 compounds) [20], and disease-focused collections such as the Chemical Genomics of Tuberculosis (47,217 molecules) [50].

Recently, extremely large chemical spaces have also been explored using on-demand combinatorial libraries such as Enamine's REAL Space (tens of billions of compounds) and its extended variant xREAL Space (trillion-scale) [73]. Unlike conventional datasets composed of explicitly enumerated compounds, these libraries are typically defined by combinatorial rules and building blocks, and therefore require specialized algorithms for efficient exploration and screening [74]. In this review, we focus on studies that operate on explicitly enumerated molecular libraries. While combinatorial design spaces are of growing importance in drug discovery, their exploration benefits from algorithms specifically designed to exploit the building-block and reaction-rule structure of these libraries, an approach that differs fundamentally from the visualization and clustering workflows reviewed here.

### Dataset scale and implications

Dataset scale constrains visualization choices. The sizes cited here correspond to those reported in the reviewed studies. Small collections, such as Heteroaromatics (45 compounds) and Amino Acids (87 compounds) [19], enable focused structural studies and permit more computationally intensive DR methods and interactive exploration. Medium-sized sets ranging from thousands to low millions of compounds, such as ZINC-DrugBank (1,490 compounds) [24], DrugBank (1,900 drugs) [29], and ChEMBL (1.56 million compounds) [35] support broader structure-activity analyses and comparative library mapping. Large to ultra-large libraries containing over one billion compounds (e.g., ZINC20 [37] and ZINC22 2D [9]) enable industrial-scale screening and diversity mapping but impose strong computational and memory demands, motivating scalable featurizations, linear or approximate DR, and clustering algorithms designed for big data. In practice, integrating datasets across these varying scales provides complementary views of diversity

**Table 2** Comparison framework overview of the 56 included studies in chemical space visualization

| Year | Preprocessing [1] | DR | Clustering | Dataset | Evaluation metrics |
|---|---|---|---|---|---|
| 2001 [18] | Spatial autocorrelation vectors; ISIS public keys (166-bit) | Multidimensional Scaling (MDS) with Neural Networks | Voronoi Partitioning (internal to DR method) | Gasteiger's combinatorial libraries (87,723 compounds); Reductive Amination Library: 90, 000 compounds; M1 Muscarinic Agonist Conformational Dataset: 5,000 conformers | Kruskal's Stress Function; Input similarity/distance measures: Tanimoto, Euclidean & Manhattan |
| 2001 [19] | Descriptor Calculation: 72 molecular descriptors | PCA | - | Trained on 423 compounds; Additional Validation Datasets: (Monocarboxylates: 8,599 structures); (Heteroaromatics: 45 compounds); (Amino Acids: 87 R-amino acids) | PCA Score Predictions; Probability of Belonging to the Model; $Q^2$; $R^2$ |
| 2003 [11] | Standard Normal Variate transformation; Baseline effect removal and warping correction | PCA; Projection Pursuit (PP); SOM; GTM; Sammon projection; Bottleneck neural network. | SOM | Data Set 1: 536 NIR spectra; Data Set 2: 83 NIR spectra of polymer products; Data Set 3: 576 objects, 159 variables | Euclidean distances; Nearest-neighbor-based projection indices |
| 2003 [20] | Molecular descriptors include physicochemical properties, structural descriptors, and pharmacophoric attributes; Representations range from lipophilicity, molecular weight, and charge to multi-point pharmacophores; Data Standardization | PCA with NIPALS Algorithm; Encoder Network-Based Nonlinear Projection | SOM | National Cancer Institute (NCI) AIDS Dataset: 29,184 compounds; World Drug Index (WDI): 5,000 compounds | Kruskal's STRESS Value; Matthews Correlation Coefficient (MCC) |
| 2004 [21] | Reduced Graphs, which abstract molecular structures by grouping chemically similar atoms into superatoms; Extended previous reduced graph models by encoding positively and negatively ionizable groups; Binary fingerprints | - | Data-Driven Clustering | NCI AIDS Screening Data (~40,000 compounds); CB1 Antagonist Dataset (~2,056,212 molecules); WDI (58,945 compounds) | Weighted Edit Distance Similarity; Daylight Similarity |
| 2006 [22] | Physicochemical properties; Data Standardization: Z-score transformation | PCA; GTM; Hierarchical-GTM (HGTM) | SOM; Gaussian Mixture Model (GMM) | Pfizer: 1 million compounds screened; 44,862 compounds screened for all five targets; 11,800 compounds selected for visualization analysis | Magnification Factor & Directional Curvature (for GTM projections); Explained Variance (PCA); KL divergence |
| 2008 [23] | Pareto scaling | PCA | Principal Component Variable Grouping (PCVG) | LC-MS metabolomics dataset | Explained Variance (%) |
| 2009 [24] | Physicochemical descriptors; Molecular ACCess System keys (MACCS); Graph-based three-point pharmacophores; Typed graph distance | PCA | - | Drugs (DrugBank from ZINC): 1,490 compounds; Natural Products (ZINC): 89,425 compounds; MLSMR: 250,000 compounds; Combinatorial Libraries: (Library I: 738, 192 compounds); (Library II: 45, 864 compounds); (Library III: 31, 320 compounds); (Library IV: 3, 552 compounds) | Nearest-Neighbor Analysis; Scaffold Analysis (Cyclic Systems) |

**Table 2** (continued)

| Year | Preprocessing [1] | DR | Clustering | Dataset | Evaluation metrics |
|---|---|---|---|---|---|
| 2013 [25] | Moreau-Broto autocorrelation descriptors | PCA | SOM | Small Molecule Universe (SMU): 8.9 million representative structures | Tanimoto Similarity Score; Principal Component Variance (%); Synthetic Accessibility Score (SA Score) |
| 2014 [26] | BCUT descriptors | - | K-Means; Bisecting K-Means; Ward's Method | NCI dataset: Six subsets of 100 to 1000 compounds each | Standard deviation of clusters; Computation time |
| 2015 [27] | Molecular Operating Environment (MOE) 2D descriptors (186 features); MACCS fingerprints (162 features); Standardization of molecular structures using Instant JChem (ChemAxon) | incremental-GTM | GTM-based Clustering | 2,193,550 molecules from 36 commercial chemical libraries and the NCI database | Shannon Entropy (for dataset diversity analysis); Tanimoto similarity coefficient (for molecular comparison); Bhattacharyya coefficient and Euclidean distance (for dataset similarity analysis) |
| 2015 [2] | Molecules encoded as SMILES strings; Molecular Quantum Numbers (MQN); SMILES Fingerprint (SMIfp); Molecules violating valency rules, steric strain constraints, and reactive functional groups were removed | PCA | - | GDB-17: 166.4 billion molecules; Filtered subsets include GDB-13 (977 million molecules) and GDB-11 (26.4 million molecules) | Tanimoto Similarity Score; Manhattan Distance in MQN Space; Principal Component Variance (%) |
| 2016 [28] | Smooth Overlap of Atomic Positions (SOAP) | Sketch-map | - | $C_{60}$ clusters: 80 local minima; Silicon: 1,274 bulk structures; Arginine dipeptide: 5,062 conformers; QM7b: 7,211 small organic molecules | MAE; RMSE |
| 2017 [29] | Extended Connectivity Fingerprints (ECFP4) for molecular representation; MACCS fingerprints used to measure 2D similarity | PCA | - | DNMT1 inhibitors (244 compounds); SMARCA2 inhibitors (220 compounds); CREBBP inhibitors (178 compounds); L3MBTL3 inhibitors (115 compounds); HDAC1 inhibitors (3,257 compounds); DrugBank (1,900 approved drugs) | Tanimoto similarity coefficient; Euclidean distance; Correlation coefficient (between full dataset visualization and satellite-based ChemMaps visualization) |
| 2017 [30] | Six different molecular fingerprints (MQN, SMIfp, APfp, Xfp, Sfp, ECFP4); Data Standardization | PCA; SIM | - | Nicotine Analogs: 5,000 compounds from GDB-13; Aurora A Kinase Inhibitors: ~1,500 compounds from ChEMBL; Aurora A Sub-Study: 48 compounds; Multi-Target Active Compounds: 197 inhibitors from the eDUD database | Principal Component Variance (%); Tanimoto Similarity Score; MST Distance |
| 2018 [31] | Molecular representations include atomic environment descriptors | - | - | ANI-1 dataset, containing 22 million conformations | Mean Absolute Error (MAE); Root Mean Squared Error (RMSE) |
| 2019 [32] | ECFP4; MACCS Keys (166-bit); Physicochemical Descriptors | PCA | Hierarchical Clustering | NuBBE: 2214 compounds | Tanimoto Coefficient; Scaled Shannon Entropy (SSE) |
| 2019 [33] | Data Standardization; Annotating scaffolds; Fragmenting: NCATS R-group; Identifying substructures: Bemis–Murcko frameworks | - | Fuzzy clustering | TCAMS dataset: 13,500 compounds; GSK kinase dataset: 9,259 compounds | Common Proportion; Proportion of Information Unique (PIU); Fragment Efficiency |

**Table 2** (continued)

| Year | Preprocessing [1] | DR | Clustering | Dataset | Evaluation metrics |
|---|---|---|---|---|---|
| 2019 [34] | Non-contiguous Atom Matching Structural similarity (NAMS), a graph-based similarity | Principal Coordinates Analysis; Kruskal MDS; Sammon Mapping; t-SNE | - | SIGMAR1: 226 compounds; HRH1: 1,208 compounds; HERG: 1,476 compounds; DRD5: 339 compounds | Area Under the Curve (AUC); MCC; Shepard Plots (for distance preservation assessment) |
| 2019 [35] | ECFP6 (2048-bit); Standardized using molvs (a Python package) | parametric t-SNE (pt-SNE) with Deep Neural Network; PCA; MDS | - | ChEMBL v23: 1,564,049 molecules; DUDe Dataset: (GPCR Ligands: 1, 480 compounds); (Nuclear Receptor Ligands: 2, 995 compounds); PubChem: 415 molecules | Classification accuracy |
| 2020 [36] | 3D conformer generation using CREST; Energy ranking of conformers using semiempirical quantum chemical calculations; Potential energy surfaces (PES); Boltzmann-weighted ensemble averaging | - | RMSD and energy-based clustering (CREST refinement) | Seven representative molecules, each explored through hundreds to thousands of conformers | mean average deviation; RMSD |
| 2020 [37] | Molecular Representations: 2D and 3D structures; Compounds are represented using SMILES strings and molecular graphs; ECFP4 | - | Bemis-Murko Scaffold | ZINC20: 1.4 billion compounds | Tanimoto Similarity Score; Graph-Edit Distance (SmallWorld Algorithm); Scaffold Diversity Analysis |
| 2020 [38] | MHFP6 (MinHash Fingerprint); ECFP4 (for comparison only) | TreeMap Projection (TMAP) | Implicit clustering via MST topology | ChEMBL: 1,159,881 compounds; FDB17 + ChEMBL: 11,261,085 compounds; DSSTox: 848,816 molecules; Natural Products Atlas: 24,594 molecules; DrugBank: 9,300 molecules; MoleculeNet benchmarks: datasets from 642 – 437,929 molecules; Other domains: Protein Data Bank (131,236 entries) | Locality Preserving Performance; Running time; Memory usage |
| 2020 [39] | Local and global structural descriptors combined with the physical properties of the different systems | DR-agnostic (sketch-map used in demonstration) | - | QM7b: 7211 molecules | - |
| 2020 [40] | MACCS Keys; SPECTRe Substructures; ChemAxon | PCA; DarkChem | K-Means; BMD cluster | ENTACT: 4346 Compounds | Silhouette Score |
| 2021 [41] | SMILES Encoding | The projection head; t-SNE | - | 2 primary datasets: 5,000 and 150,000 molecules | Tanimoto coefficient; Euclidean distance; Cosine similarity |
| 2021 [42] | SMILES-based encoding; Reaction Fingerprints (rxnfp); Transformer models generate vector representations (256-dimensional embeddings) for each reaction | TMAP | - | Pistachio Dataset: 2.6 million reactions; USPTO 1kTPL Dataset: 445,000 reactions | MCC |
| 2021 [43] | Reaction Difference Fingerprints (from ECFP, AtomPair, and Topological Torsion); BERT Fingerprints | pt-SNE | Relies on the t-SNE projections | Dataset A: US patent reactions (Lowe): ~2 million; Dataset B: Schneider 50k (20,157 after filtering) | LightGBM accuracy; Euclidean distance |
| 2021 [44] | ISIDA fragment descriptors | GTM; HGTM | Zone-based clustering | ZINC dataset: 800 million; ChEMBL: 1.6 million | Maximum Common Substructure (MCS); Density and Categorical Landscapes |

**Table 2** (continued)

| Year | Preprocessing [1] | DR | Clustering | Dataset | Evaluation metrics |
|---|---|---|---|---|---|
| 2021 [45] | ECFPs; Daylight Fingerprints; Molecular Descriptors; Mol2vec (molecular embeddings) | t-SNE; pt-SNE | t-SNE-based clustering | Serotonin-Dopamine dataset: 118 compounds; P-glycoprotein dataset: 893 compounds | Pearson and Kendall correlations (Trustworthiness Metrics) |
| 2022 [46] | ISIDA Fragment Descriptors | GTM; HGTM | Responsibility-vector-based clustering | ZINC, ChEMBL, and COCONUT libraries, from fragment-like to drug-like subspaces (15K to 516 M compounds) | Scaffold and MCS Analysis; Density Landscapes |
| 2022 [47] | Morgan fingerprints; Extended 3D Fingerprints; Lipinski's Rule of 5; 40 RDKit-derived molecular descriptors; Murcko and Schuffenhauer scaffolds; Breaking Retrosynthetically Interesting Chemical Substructures (BRICS) fragment decomposition | t-SNE; Grid-based regularization | Implicitly using t-SNE | PPDB: 560 commercial herbicides; ChEMBL: 3,130 drugs | Tanimoto similarity coefficient; ML accuracy; F1; ROC; docking scores; MM-GBSA; Mann–Whitney U tests |
| 2022 [48] | ECFP4; Physicochemical descriptors using the Mordred library | PCA; t-SNE; UMAP | K-Means | FreeSolv: 642; Huuskonen: 1291; BACE: 1513; Lipophilicity: 4200; AqSolDB: 9982; ClinTox: 1478; BBBP: 2039; HIV: 41,127 molecules | Distance Property Relationship; Compute distances |
| 2022 [49] | SMILES-based encoding; 10 physicochemical descriptors; Canonicalization, desalting, and neutralization of molecules using RDKit; Filtering to remove invalid molecules, limiting molecular weight (<1000 Da) and ensuring compatibility with the DarkChem model | PCA; DarkChem (VAE); Latent Space Encoding | - | In-house curated dataset of 2.5 billion compounds, derived from PubChem, ChEMBL, DrugBank, ECMDB, HMDB, and 35+ databases. Subset used for training and evaluation: 1,000,000 molecules after filtering | Percentage of Chemical Space Filled |
| 2022 [50] | 200 chemical descriptors derived from RDKit; 8 atomic features; 4 bond features; Z-score normalization applied to molecular descriptors | PCA; variational autoencoder (VAE) | K-Means; BIRCH | Chemical Genomics of Tuberculosis: 47,217 compounds | Silhouette Score; CHI; DBI |
| 2022 [51] | RDKit 2D fingerprint; MCS | - | Modularity-based community detection | ChEMBL: 404 compounds (459 before filtering) | Clustering coefficient; Degree assortativity; Modularity |
| 2023 [52] | Molecular Descriptors | t-SNE | Fingerprint-based Similarity Clustering | ChEMBL: 2,000 compounds; BRICS-generated: 20,000 compounds | SAS; pIC50 |
| 2023 [53] | Physicochemical descriptors; MACCS keys fingerprints (166-bit); ECFP4; Standardization of chemical structures using RDKit and MolVS; Removal of invalid molecules | t-SNE; PCA; TMAP | Implicit clustering | FooDB (Food Database): 52,856 compounds (70,477 before standardization) | - |
| 2023 [54] | Quantum-chemical descriptors | - | - | 120 DFT bulk structures per CE space; 34 selected for slab/adsorption DFT; 49 compositions synthesized experimentally | MAE |

**Table 2** (continued)

| Year | Preprocessing [1] | DR | Clustering | Dataset | Evaluation metrics |
|---|---|---|---|---|---|
| 2023 [3] | SMILES strings; 2D and 3D structures generated using CORINA and RDKit; Molecular properties: cLogP, solvation energy, atomic charges | - | Bemis-Murko Scaffold | ZINC-22: 37+ billion molecules (2D); 4.5 billion molecules in 3D | Graph-Edit Distance (SmallWorld Algorithm); Tanimoto Similarity Score; Scaffold Diversity Analysis (Bemis-Murko Scaffold Counting) |
| 2023 [55] | Normalization and Taxonomic Mapping; Data Cleaning and Filtering; Chemical Class Mapping | t-SNE | Hierarchical clustering | COCONUT+LOTUS: 87,019 unique chemicals | Szymkiewicz-Simpson coefficient; ARI |
| 2023 [56] | Standardization (ChemAxon); ISIDA fragment counts | GTM | Responsibility-vector-based clustering | ChEMBL28: 1,605,370 molecules; DNA-Encoded Libraries (DELs): 2,497 libraries (1 M representative compounds each) | Tanimoto coefficients; Coverage scores |
| 2023 [57] | Standardization; ISIDA fragment descriptors | GTM | Responsibility-vector-based clustering | ChEMBL28: 1,605,370 molecules; DELs: 2,497 libraries (1 M representative compounds each) | Tanimoto coefficient; AUC; Receiver Operating Characteristic curve (ROC) |
| 2024 [58] | MACCS, ECFP, and PubChem fingerprints | PCA | - | Approved drugs (Drugs): 1657 compounds; Nutraceuticals: 103 compounds; Mtb inhibitors I: 1223 compounds; Mtb inhibitors II: 769 compounds | Tanimoto similarity scores; Scaffold diversity; Comparing the number of unique scaffolds relative to total molecules |
| 2024 [59] | DFT-derived LKB-PPscreen ligand descriptors (28 features); StandardScaler normalization | PCA; t-SNE; UMAP | K-Means; Hierarchical Clustering | LKB-PPscreen: 275 entries | $R^2$ (Coefficient of Determination); Mean Deviation; Silhouette Score |
| 2024 [60] | SOAP | MDS; cl-MDS | K-Medoids Clustering | CHO database: 151,556 atomic environments; QM9 database: 133,885 neutral organic molecules | MDS stress; Relative intra-cluster incoherence; Euclidean distance |
| 2024 [61] | MS/MS spectra; pairwise spectral similarity via modified cosine (GNPS). RDKit fingerprints used for evaluation only | - | MNs used a novel Transitive Alignment technique | FDA Pt2, NIH SPAC, NIH NP, NIH NP Rd2 Positive, EMBL MCF, and PSU-MSMLS Datasets ranging from 476 to 7170 nodes | Network Accuracy; N20 |
| 2024 [62] | Physicochemical features computed using Dragon software; RDKit-based similarity analysis using fingerprint comparison | t-SNE | Hierarchical Clustering; Heatmap | Database of 700 organic semiconductors for OSCs; Generated Monomers: 10,000 new monomers using the BRICS method | $R^2$ score; RMSE; Synthetic Bayesian Classifier; Silhouette Score |
| 2024 [63] | Dragon molecular descriptors | PCA | - | TCMSP and DrugBank: 247 chemical compounds | Euclidean distance |
| 2024 [64] | 206 1D and 2D molecular descriptors were computed using MOE 2020.09 software; Data Standardization | PCA | Hierarchical Agglomerative Clustering with Average Linkage | Tafenoquine Analog Library: 25, 472 compounds before filtering; 1,274 compounds after applying synthetic feasibility and commercial availability filters | Space Coverage; Population Coverage; IC50 (Half-maximal Inhibitory Concentration) |
| 2024 [65] | Molecular descriptors were computed using ChemDes; Molecules were converted into SMILES format | t-SNE | Hierarchical Clustering | 300 organic semiconductors; 10,000 monomers were generated using the BRICS method in RDKit | $R^2$; RMSE; MAE; Silhouette Score |

**Table 2** (continued)

| Year | Preprocessing [1] | DR | Clustering | Dataset | Evaluation metrics |
|---|---|---|---|---|---|
| 2024 [66] | PubChem fingerprints, MACCS keys, and ECFP4; Physicochemical descriptors (6 RDKit descriptors); Standardization: Molecules were desalted, processed with RDKit, and converted to pIC50 (-log IC50) | PCA | Murcko Scaffold Analysis | ChEMBL: 3,952 molecules | Tanimoto Similarity Score; Structure-Activity Landscape Index (SALI); Scaffold Enrichment Factor; QSAR Model; Performance Metrics: Accuracy, Recall, and MCC |
| 2024 [67] | Morgan fingerprints; Scaffold-Key; Standardization | Hilbert-Curve Assisted Structure Embedding (HCASE) | - | ChEMBL v24.1 scaffolds: 55,961 unique BMSs; DrugBank: 2,073 approved drugs; CANVASS: 344 compounds | Correlation (Pearson and Kendall) between SK-rank and 2D distances |
| 2024 [68] | BRICS; ECFP4 | t-SNE | - | PI1M: 1 million polymer structures; Generated database: 50,000 polymers | Tanimoto Similarity Score; SA Score; SALI Plot |
| 2025 [8] | Morgan Count Fingerprints (radius 2, 1024-bit); MACCS Keys; ChemDist Embeddings; Standardization | PCA; t-SNE; UMAP; GTM | - | ChEMBL: 103 datasets, each containing 406 to 4,376 compounds | PNN; Trustworthiness; Continuity; Local Continuity Meta Criterion (LCMC); Scatterplot Diagnostics |
| 2025 [9] | 2048-bit RDKit fingerprints | PCA | Taylor-Butina; BitBIRCH | 30 ChEMBL target-oriented subsets; PubChem 2005 library: 1 million molecules; ZINC22 database: 1 billion molecules | CHI; DBI; DI |

[1] Preprocessing steps such as standardization are included only when explicitly reported by the original study

and coverage, and supplies realistic benchmarks for evaluating visualization pipelines.

In the next section, we describe how molecules from these sources can be featurized, since representation choices directly influence the scalability of downstream visualizations.

## Employed featurization techniques and their impact

Featurization converts molecular structures into computationally usable formats, enabling DR, clustering, and predictive modeling. In chemical space visualization, featurization methods are typically categorized as molecular descriptors and graph-based representations. Molecular descriptors comprise both continuous physicochemical and topological descriptors and discrete structural encodings such as molecular fingerprints. Each of these captures distinct views of chemical characteristics used in similarity analysis and property prediction. Figure 2 illustrates examples of these featurization techniques. In this review, we present fingerprints separately from other types of descriptors for practical clarity, as fingerprints are often treated as an independent family of representations in the literature due to their binary or count-based encoding and their prevalence in similarity search and large-scale clustering. However, fingerprints are a subtype of molecular descriptors.

This section answers **RQ2** by providing a summary of featurization techniques that have been used in the reviewed studies for chemical space visualization, outlining the recent trends and their impact on downstream tasks.

The analysis of the 56 studies presents the evolving trends of molecular featurizations in chemical space visualization. In Fig. 3, we showcase a summarizing view of these trends by counting the frequency of featurization techniques used in these studies. In the first period (2000–2010), graph-based and descriptor-based features were predominant compared to fingerprints. As molecular libraries expanded, fingerprints became the most common representation between 2011 and 2020, indicating the adoption of ECFP, PubChem, and Morgan fingerprints. Other methods (descriptors, SMILES, SOAP, quantum, and fragment-based features) saw occasional use, marking the initial steps towards hybrid pipelines. Since 2021, the field has continued to diversify: structure-based fingerprints and fragments are still dominant, followed by descriptors and SMILES encodings, then a small rise in deep learning embeddings. Generally, there is a trend away from interpretable representations toward hybrid and learned encodings, enabling efficient exploration of chemical space.

In the following subsections, we provide a general description of these featurization techniques and the studies that utilized them.

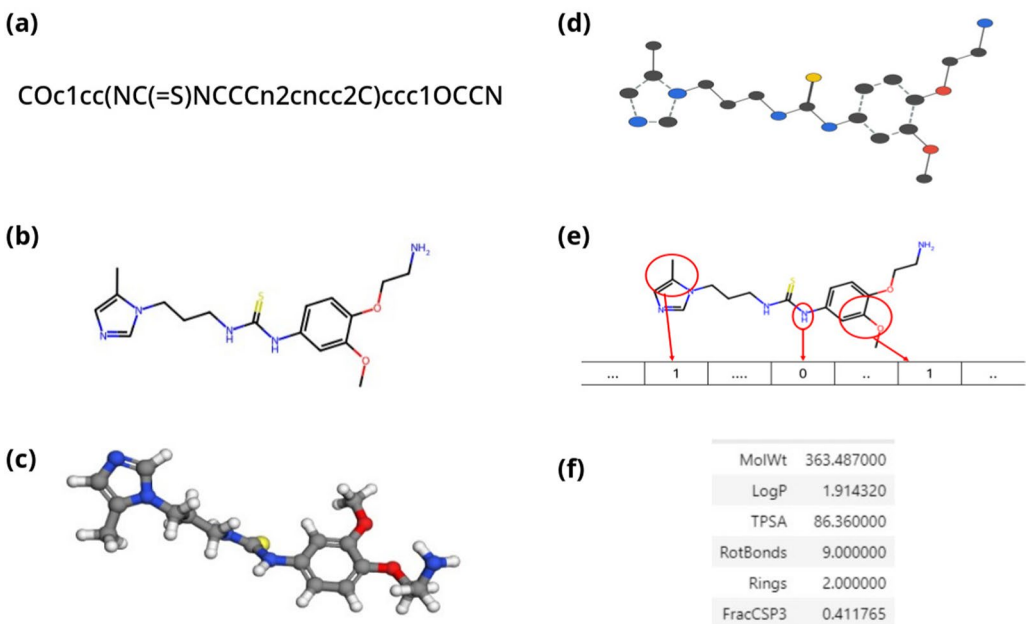



**Fig. 2** Different representations of one molecule: **a** SMILES, **b** 2D Structure, **c** 3D Conformations. Featurization Methods: **d** Graph, **e** Morgan Fingerprint, **f** Descriptors. All shown figures were generated using the RDKit package in Python [75]

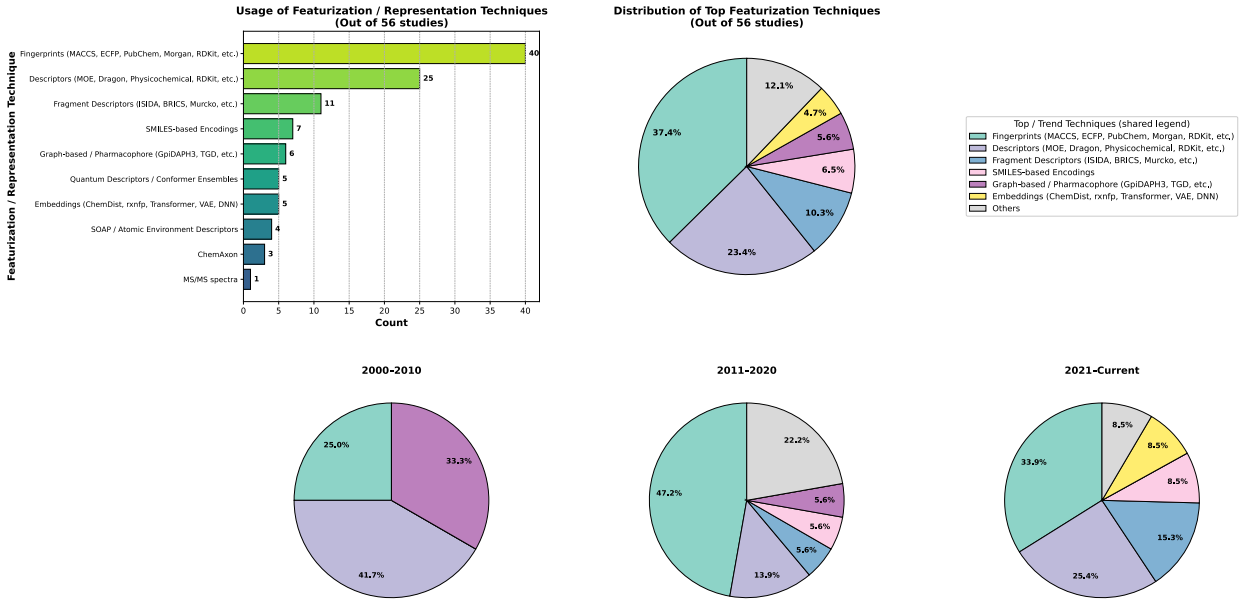



**Fig. 3** Distribution and Trends of Featurization and Molecular Representation Techniques among the reviewed studies. The total counts exceed 56 because several studies employed multiple featurization techniques

### Molecular descriptors

Molecular descriptors quantify chemical properties related to molecular size, shape, and electrostatics, translating physicochemical information into numerical form. They range from simple descriptors (e.g., molecular weight) to complex topological and 3D features. Comprehensive treatments of molecular descriptors for cheminformatics applications can be found in foundational references [76] as well as more recent work covering descriptor design and the role of descriptors in chemical space representation and computational modeling [77]. These descriptors allow for the biological interpretation of molecular interactions and play a central role in property prediction. For instance, the Chemplot study analyzing datasets such as FreeSolv, Huuskonen, and BBBP used 200 physicochemical descriptors for DR and clustering tasks [48].

### Fingerprint-based molecular descriptors

Molecular fingerprints are a subtype of molecular descriptors that transform structural characteristics into binary or integer vectors, thus indicating the presence or absence of particular molecular fragments. Among them, Morgan fingerprints provide a circular encoding framework based on atom-centered neighborhoods. They have been utilized to explore the chemical space of the ChEMBL dataset [8, 47, 67], providing adaptable encoding radii that capture both local and global features. Additionally, ECFPs [78] are widely used as an instantiation of this framework. In particular, ECFP4 has been frequently applied to datasets such as NuBBE [32] and ChEMBL [35, 66] because it effectively captures local molecular environments and structural diversity. MACCS keys (166-bit) remain popular for drug and nutraceutical datasets [29, 58], providing interpretable substructure-level representations. Such fingerprints have played a vital role in similarity search, clustering, and virtual screening.

### Graph-based representations

Graph-based techniques conceptualize molecules as graphs in which atoms are nodes and bonds are edges. Such a representation encodes the structural connectivity between atoms. Molecular graphs have been used to represent molecules from ZINC and other datasets [3, 21, 24, 34, 37, 53]. To assess molecular similarities in terms of topological patterns, graph-based similarity methods like NAMS have been used on the SIGMAR1 dataset [34]. Thus, they provide another avenue for successful structure-based learning and prediction in cheminformatics.

### Other molecular representations

Some of the molecular representations that have been employed in the literature do not belong to the above-mentioned categories. These alternative approaches comprise string-based encodings and learned embeddings. SMILES strings provide a one-dimensional textual notation of chemical structures and have been used for dataset preprocessing, visualization, and deep learning inputs [2, 3, 25,

37, 41, 42, 49, 59, 65]. Although molecular graphs can be reconstructed from SMILES, the encoding itself is sequential rather than graph-structured. Representation-learning methods also produce dense continuous embeddings of molecules or reactions, such as rxnfp and transformer-derived vector encodings [8, 42, 48].

These alternative representations go beyond the classical cheminformatics types to allow flexible, model-driven featurization of molecules that can be used to complement fingerprints, descriptors, and graph-based methods.

#### Combination of featurization techniques

Several studies have employed hybrid featurization techniques that combine the strengths of diverse methods to complement each other. For example, a combination of ECFP, MACCS, and descriptor features enhanced clustering performance and diversity analysis in Approved Drugs, Nutraceuticals, and Mtb inhibitor datasets [58]. Similarly, the combination of ECFP fingerprints and ISIDA descriptors has been applied to ChEMBL28 and DEL datasets, enabling multi-scale structural mapping [57]. Another study incorporated SMILES strings, Morgan fingerprints, and RDKit descriptors into the PPDB and ChEMBL datasets, thus generating detailed molecular representations, which later enhanced predictive modeling and similarity evaluation [47]. These hybrid methods produce more substantial representations, thereby improving the interpretability and robustness of chemical space analysis.

#### Impact of featurization on downstream tasks

The performance of DR, clustering, and predictive modeling is directly affected by the quality of featurization. Combining various techniques such as ECFP and MACCS with PubChem fingerprints yields superior clustering and visualization, as reported by Lin et al. [79], thereby enhancing molecular mapping. The combination of descriptors and fingerprints enhances virtual screening and similarity searching by capturing complementary structural information [80]. In addition, predictive models gain an advantage from having hybrid feature sets consisting of descriptors, fingerprints, and graph-based encodings, which together improve the prediction of bioactivity and toxicity [81]. In line with this, the incorporation of multiple featurization methods can have significant effects on achieving high-performance molecular analytics in chemical space studies.

## Dimensionality reduction in chemical space visualization

DR techniques convert high-dimensional molecular data into lower dimensions, particularly 2D and 3D for visualization purposes. The quality of the reduction and corresponding visualization in capturing structural relationships between molecules provides promising assistance for computational chemists in drug discovery and molecular modeling efforts. Figure 4 illustrates an example of popular DR techniques on a toy dataset showcasing the reduction from 3D to 2D. This section responds to **RQ3** by exploring the evolution of DR methods used for chemical space visualization over the selected period, outlining the recent trends, limitations, and complexities.

Most DR techniques discussed in this review, including PCA, t-SNE, UMAP, SOM, and GTM, are generally applied in an unsupervised manner for chemical space visualization, where the primary objective is to discover patterns of structural similarity without labeled outcomes. However, several of these techniques have supervised or semi-supervised variants (e.g., supervised PCA, supervised UMAP), and others can be integrated into supervised workflows in which class labels influence the structure of the latent space [82–85].

Across the reviewed studies, DR is predominantly used to visualize high-dimensional molecular data by projecting it into 2D or 3D spaces, which facilitates analysis of chemical similarity, structural diversity, and clustering relationships [32, 34, 35, 45, 48, 53, 56–58, 62, 66, 68]. Additionally, several studies have used visualization techniques as secondary analysis tools to identify activity cliffs, structure-activity relationships, or interesting clusters of active molecules for drug development and material synthesis [52, 53, 55, 62, 66, 68]. Some studies have also explored the use of several DR techniques (e.g., PCA, t-SNE, UMAP, GTM) comparatively for mapping quality and interpretability [8, 35, 59].

We compiled the distribution of DR techniques in the reviewed studies in Fig. 5. The distributions provide insight into how preferences have changed over time. In the earliest period (2000–2010), PCA dominates, accounting for roughly 38% of reported uses, with GTM and MDS appearing sporadically and the remainder spread across various other methods. Between 2011 and 2020, the dominance of PCA stabilizes (36.8%) and GTM usage drops, while non-linear and topology-based approaches gain traction, including t-SNE/pt-SNE, MDS family methods, and the first appearances of TMAP. In the most recent period (2021–present), PCA usage declines (24.4%), t-SNE-based methods rise to become dominant, GTM resurges, UMAP appears, and TMAP continues to be used, whereas MDS usage decreases. Overall, these trends indicate a gradual shift from PCA-centric practice toward a mixed portfolio of manifold-learning and graph-based DR methods.

In the following subsections, we discuss several families of commonly used techniques for chemical space visualization. These include linear projection techniques such as PCA; non-linear techniques encompassing

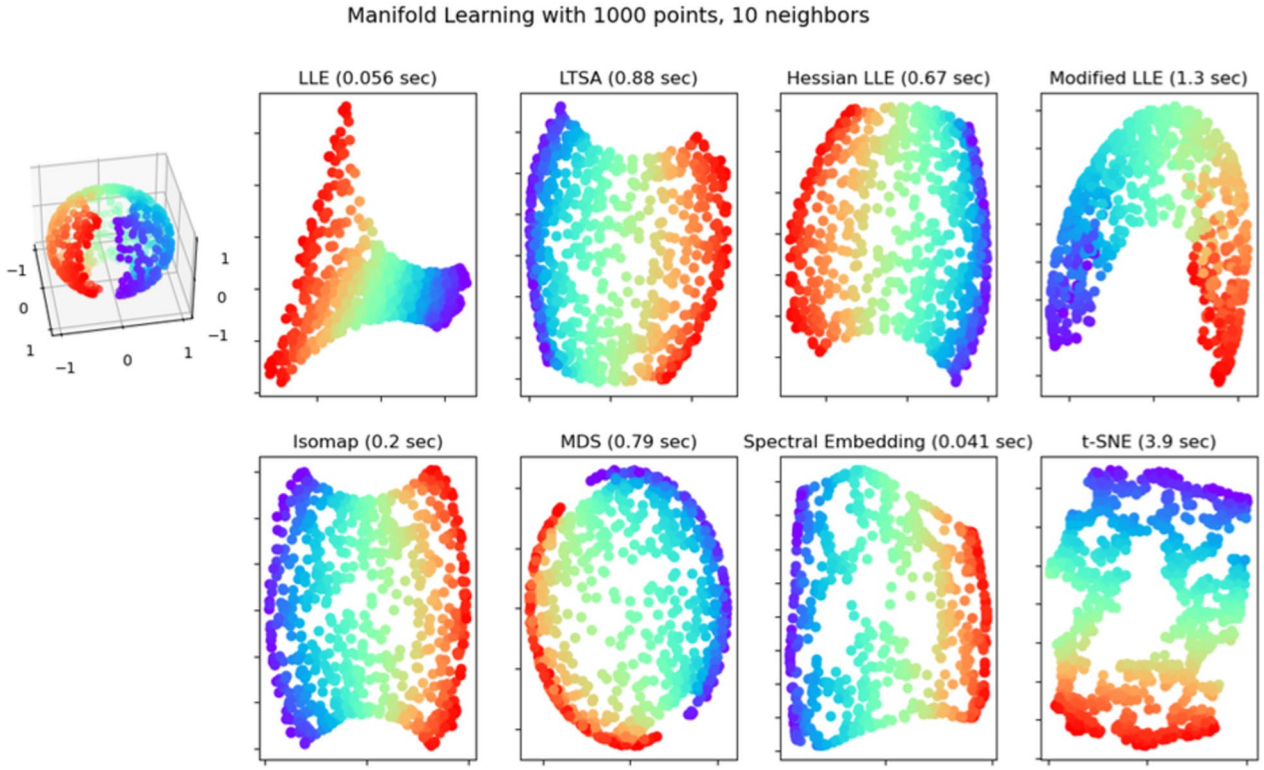



**Fig. 4** An example showcasing popular DR techniques reducing a dataset from 3D to 2D. This figure is from scikit-learn [86]

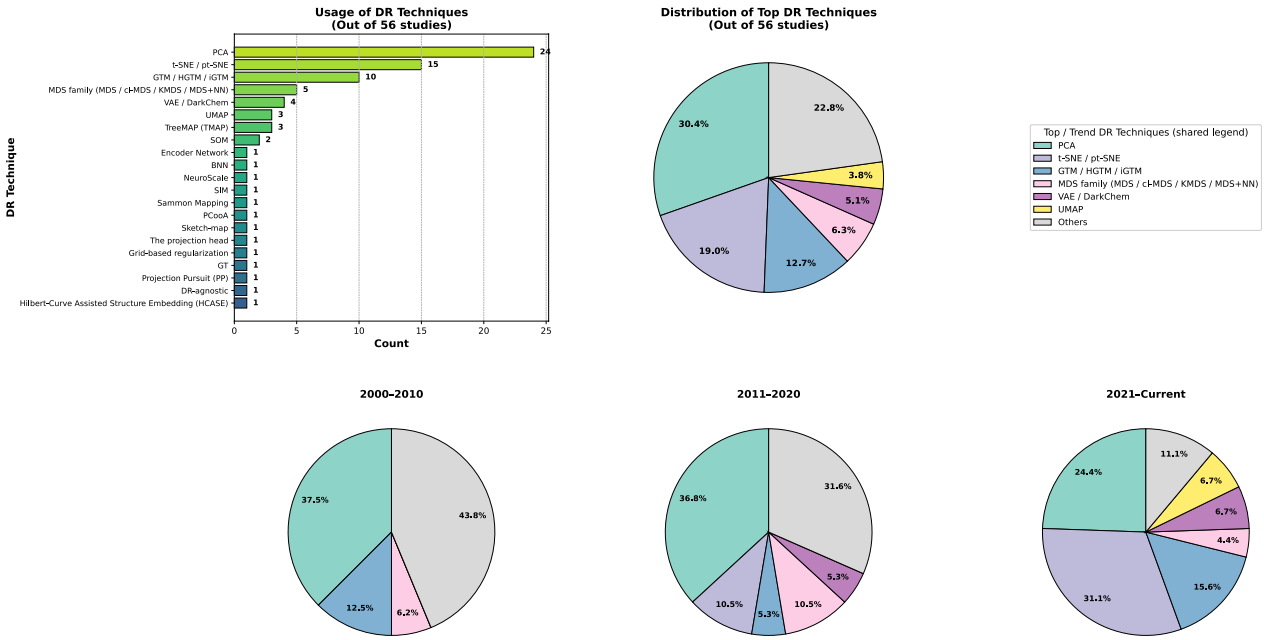



**Fig. 5** Distribution and trends of DR techniques in the 56 studies. The total counts exceed the total number of studies because many analyses applied multiple reduction techniques for comparison

manifold learning methods (t-SNE and its parametric variant pt-SNE, UMAP), distance-preserving methods (MDS), and graph-based layouts (TMAP); and latent variable models such as GTM and VAE. These methods and their scalability are discussed in subsections Overview of key reviewed DR techniques and Scalable DR for large datasets.

#### Overview of key reviewed DR techniques

PCA projects data into lower dimensions while retaining maximum variance [87]. It is fast, interpretable, and effective for large datasets like ChEMBL and ZINC [88]. However, its linear assumption limits performance with complex non-linear molecular features, and PCA depends heavily on descriptor quality [89–92].

t-SNE, UMAP, TMAP, and MDS are better suited for non-linear molecular manifolds. t-SNE excels at preserving local similarities and identifying clusters in datasets like FreeSolv [93, 94], though its complexity makes it slow for ultra-large libraries [95, 96]. Parametric t-SNE (pt-SNE) utilizes deep learning to approximate t-SNE projections, offering a scalable, out-of-sample extension for new compounds while retaining local structural fidelity [97–99]. UMAP offers a more scalable alternative, maintaining both local and global structures [84, 100], but requires careful hyperparameter tuning to avoid misleading visualizations [101, 102]. TMAP represents the chemical space as a graph-based layout based on approximate nearest-neighbor graphs and minimum spanning trees, providing exceptional scalability and preserving the local and global structures of ultra-large datasets, improving upon the visualization clarity of previous methods while lacking a reusable parametric mapping [38]. MDS preserves global distances for faithful similarity mapping [103] but is computationally prohibitive and highly sensitive to the chosen distance metric [104, 105].

GTM provides a probabilistic mapping that reveals latent structures. Beyond projection, the latent grid nodes of GTM may also be interpreted as soft cluster centers using responsibility vectors: each latent point identifies a region of descriptor space, allowing the map to be used for both dimensionality reduction and clustering simultaneously. Latent nodes may further be grouped into superclusters on the basis of chemical property distributions or density patterns [106–108]. SOM similarly combines DR and clustering in a single framework and is discussed in subsection Partitional, soft, and hierarchical clustering. VAEs learn non-linear latent spaces optimized for de novo molecular generation [109, 110], but require extensive training iterations $T$ and suffer from high parameter-tuning overhead [111, 112].

In addition to the widely adopted DR techniques discussed above, domain-specific approaches designed for cheminformatics have also been proposed. HCASE maps molecular structures into a low-dimensional space using Hilbert space-filling curves and scaffold-key representations, offering deterministic embedding with interpretable neighborhood preservation [67]. As a recently introduced method, its broader adoption remains to be seen across diverse chemical datasets.

Table 3 summarizes the strengths, weaknesses, and computational costs of these techniques.

#### Scalable DR for large datasets

PCA remains a baseline for efficiency; by utilizing truncated implementations for 2D projections, it achieves $O(nd)$ complexity, facilitating the analysis of million-scale molecular datasets on standard hardware. While t-SNE effectively captures local structures, its iterative phase is computationally expensive even with Barnes-Hut approximations, as it requires $O(n \log n)$ operations per iteration [118, 119]. Notably, parametric t-SNE does *not* require a one-time $O(n^2)$ similarity matrix over the full dataset; instead, it computes pairwise probabilities within mini-batches, giving quadratic cost in terms of the batch size. Accordingly, its training complexity is commonly expressed as $O(Tn(Bd + LH^2 + Hd))$ with working memory $O(LH^2 + B^2 + B(H + d))$, where $B$ is the batch size, while out-of-sample inference is a single forward pass through the network [126]. In contrast, UMAP improves scalability by separating approximate $k$-NN graph construction, typically $O(n \log n)$, from graph-based stochastic optimization with per-iteration cost $O(nk)$, giving an overall complexity of $O(n \log n + Tnk)$ [102]. This allows UMAP to maintain both local and global structural integrity while handling large-scale chemical spaces more efficiently.

Latent variable models such as GTM and VAE provide expressive non-linear mappings, but their training cost can be substantial due to iterative optimization and model size. For GTM, training scales as $O(T(ngd + b^3))$ with memory $O(ng + bd)$, where $g$ is the number of latent grid points and $b$ is the number of basis functions [129]. For VAEs, training is typically linear in the number of samples, $O(Tn(LH^2 + Hd))$, with working memory dominated by model parameters and batch activations (rather than pairwise structures) [131].

TMAP targets extreme-scale visualization by avoiding iterative embedding optimization. It constructs a $c$-approximate $k$-NN graph using an augmented LSH forest query, with overall time complexity $O(n(kk_c + \log n))$ (where $k_c$ is the query augmentation factor) and memory $O(nk)$, and then performs a minimum spanning tree construction whose cost is negligible relative to graph construction in practice [38]. Benchmarks further report empirical sub-linear scaling of approximately $O(n^{0.931})$ on

**Table 3** DR techniques analysis

| Category | DR technique | Strengths | Limitations | Time Complexity [1] [2] | Space Complexity [1] [2] |
|---|---|---|---|---|---|
| Linear | PCA [2, 8, 9, 11, 19, 20, 22–25, 29, 30, 32, 35, 40, 48–50, 53, 58, 59, 63, 64, 66] | • Computationally efficient and scalable for large chemical databases [88].• Preserves maximum data variance while reducing dimensionality and simplifying complex datasets [87].• Provides interpretable outputs; principal components reflect directions of variance, aiding pattern discovery and cluster analysis [87].• Familiar and widely used approach, offering a clear starting point for exploratory chemical space analysis [87] | • Limited ability to capture non-linear molecular relationships [89, 90, 113].• Strongly dependent on the quality and relevance of molecular descriptors [91].• Fails to fully represent complex or non-linear regions of chemical space [92] | Truncated PCA vs full PCA gives $O(ndc)$ and $O(nd^2 + d^3)$, respectively, where $c$ is the number of top principal components retained. [114] | Truncated PCA vs full PCA gives $O(cn + cd)$ and $O(d^2)$, respectively [114] |
| Non-Linear | t-SNE [8, 34, 41, 45, 47, 48, 52, 53, 55, 59, 62, 65, 68] | • Effectively preserves local structures in high-dimensional data [115].• Helps identify clusters of structurally similar molecules, making it suitable for analyzing compound families in chemical space [93].• Performs well on complex datasets like FreeSolv and ChEMBL, where subtle structural differences matter [94].• Produces intuitive 2D and 3D visual representations that facilitate exploration of chemical space [35].• Aids in pattern recognition by enabling straightforward visual identification of molecular relationships [43] | • Computationally intensive due to its reliance on all pairwise distance calculations [95].• Produces non-deterministic results across runs, reducing reproducibility in visualizations [96].• May distort the global geometry of the data despite preserving local relationships [116].• Can fail to represent the overall structure of chemical space, limiting insight into broader molecular relationships [117] | $O(dn \log n + Tn \log n)$ using approximations; $O(dn^2 + Tn^2)$ with naive implementation [118, 119] | $O(nk)$ using approximations; $O(n^2)$ with naive implementation [118, 119] |
| Non-Linear | UMAP [8, 48, 59] | • Maintains both local and global structures of chemical space effectively [120].• Offers higher computational speed than t-SNE, enabling better scalability for large datasets [84].• Efficient in preserving structure while delivering faster performance [100].• Generates clear 2D and 3D visualizations without compromising molecular relationship interpretation [8]. • Better at preserving full chemical space geometry compared to t-SNE [121] | • Highly sensitive to hyperparameter settings; suboptimal tuning can lead to misleading visualizations [101]. • Projection outcomes are influenced by preprocessing choices and algorithmic design [122].• Less reliable for small datasets; performance can degrade in low-data scenarios [102] | $O(n \log n + Tnk)$ using approximate $k$-nearest neighbor; $O(dn^2)$ with naive exact neighbor search [102] | $O(nk)$ [102] |

**Table 3** (continued)

| Category | DR technique | Strengths | Limitations | Time Complexity [1] [2] | Space Complexity [1] [2] |
|---|---|---|---|---|---|
| Non-Linear | MDS [18, 34, 35, 60] | • Preserves relative distances between data points, making it well-suited for chemical space exploration [123].• Strong at maintaining the global structure of data, unlike methods such as t-SNE [103, 124]. • Particularly effective when similarity is expressed through distance-based metrics [125] | • Requires substantial computational resources, limiting its efficiency on large datasets [104].• Highly sensitive to the choice of distance metric, which can affect the reliability of results [105] | $O(n^3)$ [104] | $O(n^2)$ [104] |
| Non-Linear | TMAP [38, 42, 53] | • Highly scalable, capable of visualizing billions of data points through efficient hashing and tree layouts [38]. • Preserves global topology better than t-SNE by using an MST [38]. • Ideal for high-throughput screening and massive chemical databases | • Does not produce a continuous, reusable embedding function [38]. • Visual output is restricted to a tree-like structure, which may oversimplify complex clusters. • No out-of-sample extension for new points without re-indexing | $O(n(kk_c + \log n))$ using LSH and Kruskal's algorithm for MST construction.Empirically shown to be $O(n^0.931)$. [38] | $O(nk)$ [38] |
| Non-Linear | pt-SNE [35, 43, 45] | • Improves computational speed over traditional t-SNE, making it more practical for large datasets [98].• Utilizes deep learning to handle larger input sizes while preserving local structure [99].• Maintains strong performance in preserving local chemical information during projection [43].• Designed to enhance scalability and efficiency compared to conventional t-SNE [97] | • Requires careful training and hyperparameter tuning, which adds complexity [97].• Inherits core limitations of t-SNE, including probabilistic variability and poor global structure preservation [98].• Still computationally expensive due to its reliance on neural networks [97] | $O(Tn(Bd + LH^2 + Hd))$ [126] | $O(LH^2 + B^2 + B(H + d))$ |
| Latent Variable Models | GTM [8, 11, 22, 27, 44, 46, 56, 57] | • Implements probabilistic mapping that captures data uncertainties in complex, high-dimensional datasets using a predefined latent grid and Gaussian mixture model [107].• Offers hierarchical and structured visual representations, useful for understanding relationships across multi-scale chemical spaces [106].• Supports DR via learned latent structures, allowing data exploration and generation [127]. • Simultaneously enables projection and soft clustering via responsibility vectors | • Requires substantial computational resources, especially for large-scale datasets [108].• Tuning parameters to capture accurate latent spaces is complex and difficult [128] | $O(T(ngd + b^3))$ for probabilistic latent grid optimization [129] | $O(ng + bd)$ [129] |

**Table 3** (continued)

| Category | DR technique | Strengths | Limitations | Time Complexity [1] [2] | Space Complexity [1] [2] |
|---|---|---|---|---|---|
| Latent Variable Models | VAE [49, 50] | • Transforms complex, high-dimensional molecular data into meaningful latent representations for generation and analysis [130].• Effectively captures non-linear relationships, making it well-suited for molecular datasets [109].• Enables the generation of new molecular structures through its neural latent representation, supporting de novo molecule design [110] | • Requires significant computational resources and time-consuming training on large datasets [111].• Highly sensitive to hyperparameter settings, with poor tuning leading to overfitting or suboptimal performance [112] | $O(Tn(LH^2 + Hd))$ for neural network encoder–decoder training [131] | $O(LH^2 + H(d + z))$ [131] |

[1] Definitions: $n$: Number of data points

$d$: Number of features

$c$: Number of extracted components ($c \leq d$)

$L$: Number of layers in the encoder/decoder

$k$: Number of neighbors

$H$: Number of hidden units

$z$: Number of features in the latent space

$g$: Number of latent grid points

$b$: Number of basis functions

$T$: Number of iterations

$k_c$: Query augmentation factor

$B$: Batch size

[2] Complexity not explicitly reported in the cited source was derived from our analysis of the algorithm as described in the referenced paper

large chemical databases [38]. However, TMAP does not learn a reusable parametric mapping, since it produces a static tree layout optimized for visualization.

Overall, for million-scale molecular datasets, truncated PCA provides a fast linear baseline, while UMAP and TMAP offer near-linear-to-linearithmic $O(n \log n)$ scaling that supports high-throughput exploratory analysis. By contrast, methods with quadratic (or worse) dependence on $n$ are typically confined to smaller samples or require approximations. Parametric models such as VAEs and pt-SNE are preferred when a reusable embedding function is required for out-of-sample prediction, but their training overhead and constant factors can be limiting at ultra-large scale [126, 131].

The findings shown in this section contribute to answering **RQ5** by illustrating the scalable DR techniques employed in the reviewed studies for large chemical datasets and highlighting their computational trade-offs.

## Clustering algorithms used for chemically meaningful groupings and quality evaluation

Clustering remains central in chemical space visualization, particularly to group structurally similar molecules to reveal hidden patterns [132–134]. Figure 6 shows an example of clustering 50,000 chemicals using VAE + K-means, forming 50 distinct clusters that represent chemical diversity, where t-SNE is used to visualize the resulting clusters [50]. In this example, clustering is the primary analytical task, and t-SNE serves to illustrate the structure of the clustering results rather than to define the chemical space itself. While clustering can function as a standalone analytical method, in the context of chemical space visualization it is commonly used to identify structurally coherent regions within the mapped space. In some workflows, clustering is the main objective and techniques such as t-SNE are used solely to display the resulting groups. In others, clustering is applied within low-dimensional embeddings or descriptor spaces to aid interpretation. It is therefore important to distinguish between workflows where clustering is conducted independently and where it directly supports or is integrated with visualization, as in SOM and GTM. In this review, clustering methods are discussed primarily in the context of supporting chemical space visualization and interpretation, where their role is to facilitate the identification of structural patterns within the mapped chemical space. This section answers **RQ4** by summarizing the applied clustering algorithms for chemical space

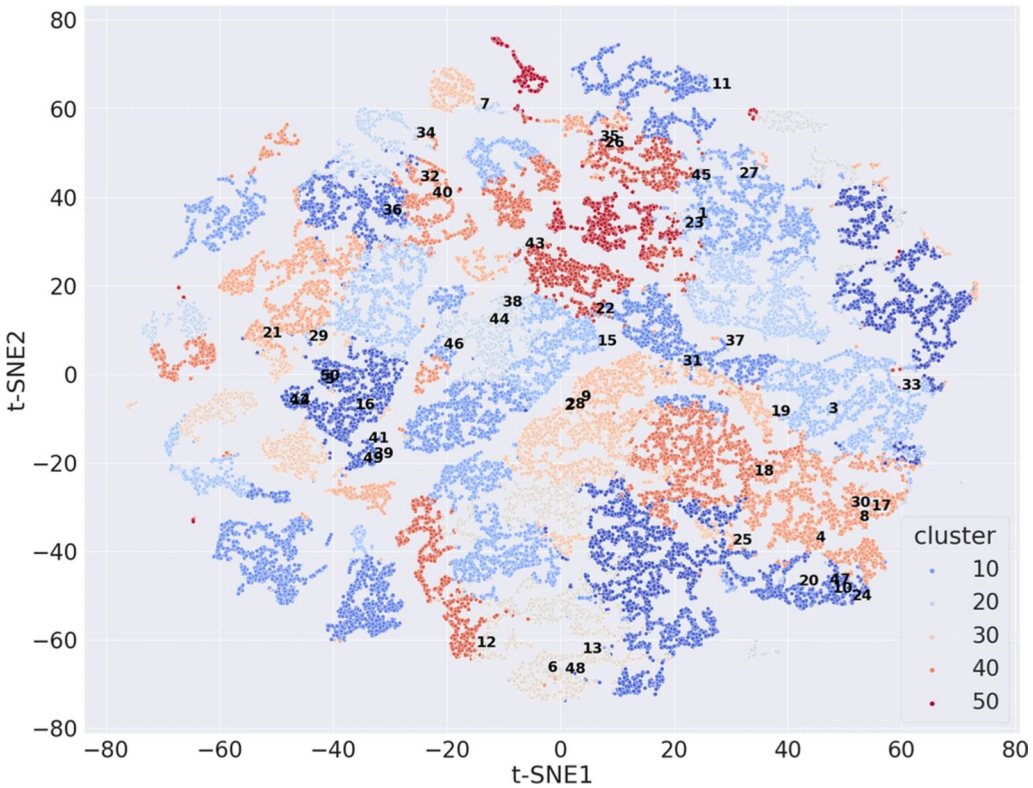



**Fig. 6** Example of clustering 50,000 chemicals into 50 clusters. Numbers indicate cluster IDs from VAE + K-means, and colors distinguish the clusters in the t-SNE visualization. The main goal of this clustering is to assess and ensure the chemical diversity of the molecular dataset. This figure is from Hadipour et al. [50] under the CC BY-NC-ND 4.0 license (http://creativecommons.org/licenses/by-nc-nd/4.0/)

visualization, the evolution of their trends over time, outlining their strengths and limitations, their complexities, and describing the evaluation metrics applied to assess results quality.

The clustering methods applied in chemical space analysis are predominantly employed in an unsupervised manner to group structurally similar molecules. Algorithms such as K-means, hierarchical clustering, SOM, and density-based methods are typically applied without labeled outcomes. However, some approaches such as K-means and SOM can be incorporated into supervised or semi-supervised frameworks by introducing label information, which can alter the resulting groupings [85, 135].

### Clustering algorithms

The clustering methods reviewed in this paper are presented in the context of chemical space visualization, where they are used to identify structurally coherent regions and to support the interpretation of molecular similarity relationships. The reviewed studies reveal that while a wide range of clustering algorithms have been explored, a small number of method families recur most frequently, alongside a substantial long tail of less commonly used approaches (Fig. 7). During 2000–2010, SOM and K-means were the most frequently recurring techniques, while many other methods appeared sporadically. Between 2011 and 2020, K-means and hierarchical clustering were the most commonly reported named methods: K-means remained attractive for its practical scalability in large libraries, whereas hierarchical clustering was often used for its interpretability (e.g., dendrogram-based, multi-resolution inspection) and in settings where quadratic pairwise costs were manageable (e.g., reduced representations or subsets). Across both periods, additional approaches such as fuzzy, scaffold-based, and density-driven variants were used intermittently, reflecting methodological diversification in chemical data analysis.

From 2021 to the present, method choice further diversified with the growth of large and complex molecular datasets. Hierarchical clustering remains the most frequent named family, while K-means, Bemis-Murcko scaffold grouping, and responsibility-vector-based methods appear with comparable frequency. Domain-specific similarity-threshold techniques such as Taylor-Butina can yield structurally coherent clusters when the cutoff is well tuned, but their reliance on pairwise similarities makes them difficult to apply at million-to-billion scale without approximations or batching. Bemis-Murcko scaffolds provide a standard abstraction that retains ring systems and connecting linkers while removing peripheral substituents, and such scaffold summaries have been applied at very large scale (e.g., the 4.5B subset of ZINC-22) [3]. Other algorithms such as responsibility-vector-based, BIRCH/BitBIRCH, and probabilistic clustering schemes are less common but show promising results. Overall, the temporal development reflects a shift from early reliance

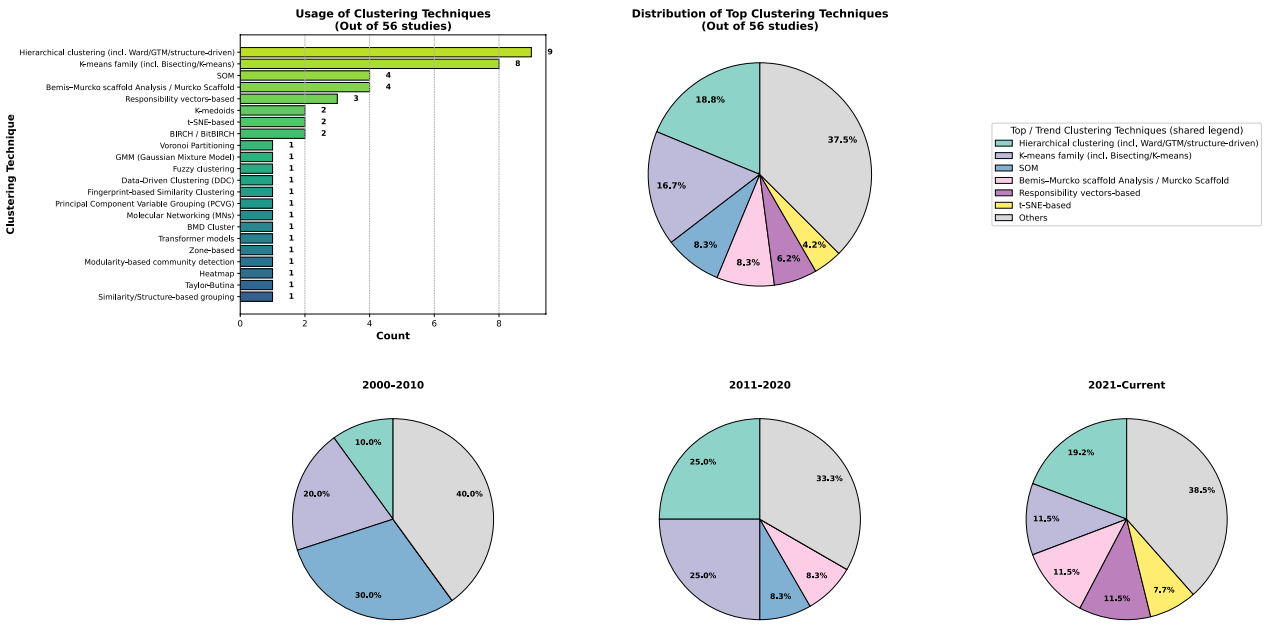



**Fig. 7** Distribution and trends of clustering techniques in the 56 studies. The total counts exceed the number of reviewed studies, as several papers used more than one clustering method

on a few standard algorithms toward more diverse and comparative evaluations aimed at improving interpretability and robustness of chemical space groupings.

We can categorize these clustering algorithms into partitional, including K-means, K-medoids, SOM, BIRCH, Taylor-Butina, and Bemis-Murcko scaffold; hierarchical, including agglomerative hierarchical clustering; and soft clustering, including fuzzy and responsibility-vector-based clustering (Fig. 8). Brief descriptions of these techniques are provided in the following subsection Partitional, soft, and hierarchical clustering.

#### *Partitional, soft, and hierarchical clustering*

Clustering methods used in chemical space analysis can be grouped into partitional (hard assignment), soft (probabilistic/fuzzy assignment), and hierarchical (multi-resolution) families. Across these categories, the choice of molecular representation (e.g., continuous descriptors vs. binary fingerprints) and the associated dissimilarity measure (e.g., Euclidean vs. Tanimoto) strongly influence the resulting clusters and their interpretability [136, 137].

Partitional clustering assigns each molecule to exactly one cluster based on an optimization objective or a rule-based partition. Centroid-based approaches such as K-means minimize within-cluster dispersion in a chosen feature space [138, 139], whereas K-medoids uses representative data points (medoids) as cluster centers, which can improve robustness to atypical observations depending on the dissimilarity definition [140, 141]. In chemical space visualization, such methods are frequently used with low-dimensional embeddings in order to detect cluster boundaries and allow their visual interpretation.

Prototype-based neural approaches such as SOM assign each molecule to a best-matching unit on a 2D lattice of reference vectors. Although SOM is commonly applied as a clustering method, it also functions as a topology-preserving DR technique by mapping high-dimensional data onto a structured 2D grid. Each unit of the map can be considered a cluster, and neighboring

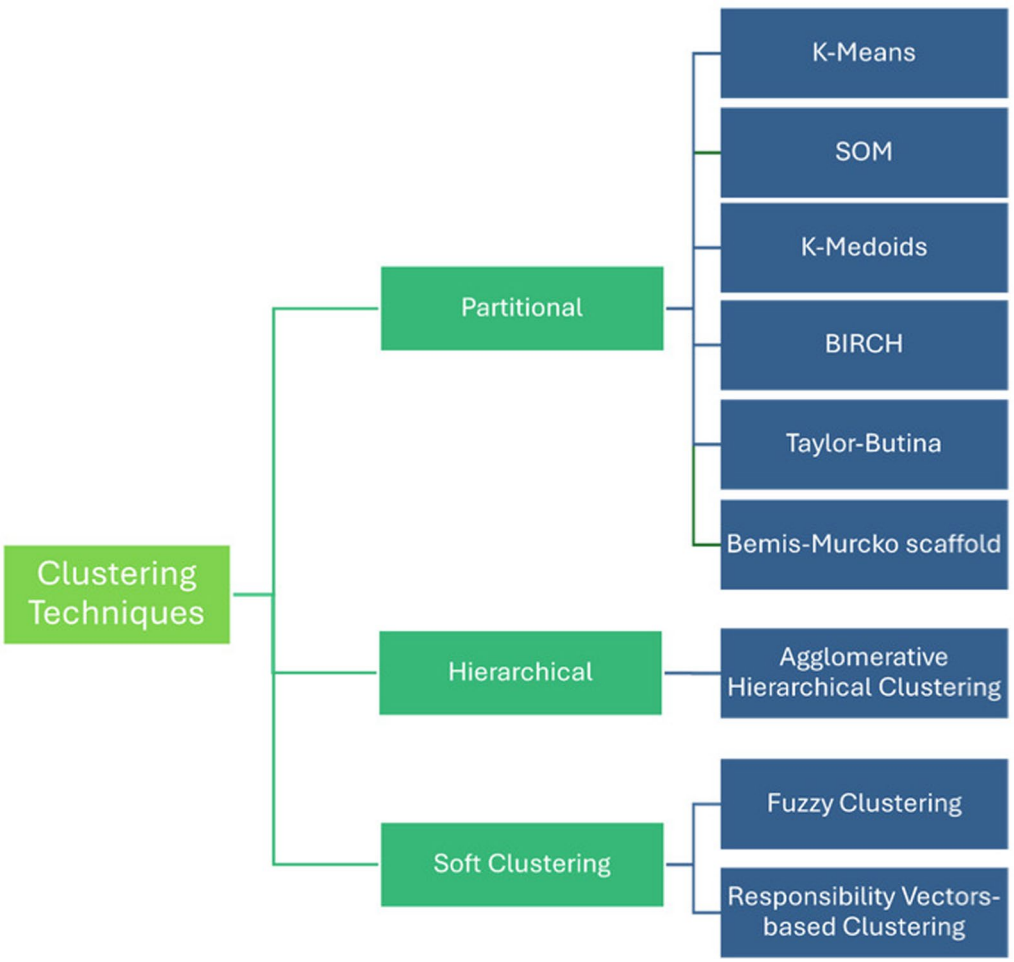



**Fig. 8** Taxonomy of clustering techniques used in chemical space analysis from the 56 reviewed studies

units maintain structural continuity, which enables DR and clustering to be conducted in a single framework [11, 20, 22, 25, 142]. In this case, clustering is closely integrated with the visualization, since the map is composed of discrete organized units representing prototypical regions of the projected space. BIRCH introduces an incremental summarization strategy via clustering features (CF) and a CF-tree, enabling hierarchical compression of the dataset into compact micro-clusters that can be further refined by downstream procedures [143]. Similarity-threshold clustering is common in cheminformatics: the Taylor-Butina procedure forms clusters by selecting representative centroids in a fingerprint similarity graph induced by a user-defined cutoff (rather than specifying a target number of clusters a priori) [144–146].

Other methods perform deterministic structural partitioning based on predefined chemical abstractions rather than on distances in descriptor or latent space [147]. Bemis-Murcko scaffold grouping assigns each molecule to a structural class by extracting its molecular framework: all atoms participating in ring systems and the linker atoms connecting them are retained, while peripheral substituents are removed. The resulting framework can be further abstracted to a generic scaffold by converting all atoms to carbon and all bonds to single bonds, yielding an unlabeled graph that defines groups solely on the basis of core topology. Because this grouping criterion is derived directly from the molecular graph rather than from any descriptor or latent representation, scaffold-based partitioning is independent of the featurization and DR choices made elsewhere in the pipeline, and is therefore better understood as structural classification than as distance-based clustering. This property makes it particularly useful for scaffold diversity analysis and large-scale library characterization [37, 148, 149]. This approach offers structure-driven grouping that directly facilitates the interpretation of chemical space through the visual organization of chemotypes.

Several scaffold-based visualization approaches incorporate structural grouping directly into the visualization process. For example, Scaffold Hunter organizes chemical space using hierarchical scaffold trees, in which molecules are grouped according to shared structural frameworks and visualized through an interactive tree representation annotated with bioactivity information [150]. In this approach, clustering is built into the visualization through scaffold-based partitioning rather than applied as a separate step. Similarly, constellation plots combine scaffold-based grouping with coordinate-based chemical space representations. Molecules are first grouped into analog series defined by shared core structures, and these cores are then displayed in a coordinate-based low-dimensional representation [151]. Each point in the visualization represents a core rather than an individual molecule, and connections between points reflect shared compounds across analog series. This integration of structural grouping and visualization enables the identification of structure-activity relationships directly within the projected chemical space.

Soft clustering assigns partial memberships, allowing a molecule to belong to multiple clusters with different degrees of affiliation. Fuzzy c-means is a canonical example, producing a membership matrix that can capture overlapping chemical series or ambiguous boundaries between chemotypes [152–154]. Responsibility-vector approaches (including GTM-like responsibility assignments) similarly represent membership through probabilistic responsibilities over latent nodes or components, which can be advantageous when uncertainty quantification or mixed-membership structure is of interest [155, 156].

Hierarchical clustering organizes molecules into a nested structure (dendrogram) through iterative merges (agglomerative) under a chosen linkage criterion and dissimilarity measure [157–159]. This representation supports multi-resolution inspection of chemical series by choosing a cut level post hoc, and it is often used to interrogate relationships between clusters, scaffolds, or representative compounds [160, 161].

Table 4 summarizes strengths, weaknesses, and computational costs of these techniques. The scalability of these techniques is provided in the following subsection Scalable clustering techniques for large datasets.

#### Scalable clustering techniques for large datasets

Building on subsection Scalable DR for large datasets, this subsection continues to answer **RQ5** by focusing on scalable computational strategies for clustering large-scale chemical datasets, highlighting algorithms and implementation-level optimizations that enable efficient grouping of millions to billions of molecules.

Scalability and memory efficiency are critical issues when clustering multi-millions or billion-scale molecular libraries. A key limitation is the quadratic barrier induced by pairwise comparisons: agglomerative hierarchical clustering typically requires $O(n^2)$ memory (e.g., storing a dense pairwise distance matrix) and ranges from $O(n^2)$ to $O(n^3)$ time depending on linkage and implementation, making direct application to ultra-large libraries prohibitive. Similarly, Taylor-Butina clustering relies on computing all-pairs fingerprint similarities, incurring $O(n^2)$ time and space, which quickly becomes infeasible at very large $n$ [9, 144]. This limitation extends to K-medoids, which is typically quadratic in memory (distance matrix) and approximately $O(Tkn^2)$ in time (after

**Table 4** Clustering techniques analysis

| Category | Clustering technique | Strengths | Limitations | Time Complexity[1 2] | Space Complexity[1 2] |
|---|---|---|---|---|---|
| Partitional | K-Means [26, 40, 48, 50, 59] | • Efficient for large datasets such as ZINC and ChEMBL due to its computational simplicity [162]. • Widely used in chemical compounds grouping; operates using distance-based similarity (e.g., Tanimoto coefficient) [136].• Performs well in grouping structurally and chemically distinct compounds with clear cluster boundaries [163]. • Quick to implement and yields fast clustering results, even with minimal configuration [139] | • Highly sensitive to the choice of initial centroids, which can lead to inconsistent or suboptimal results [164]. • Requires users to specify the number of clusters (K) in advance, which is difficult in chemical space analysis [165]. • Relies on distance metrics like Euclidean distance, making it less effective for non-spherical or overlapping cluster shapes [166] | $O(Tnkd)$ [167] | $O(kd + n)$ [167] |
| | SOM [11, 20, 22, 25] | • Produces a topology-preserving grid of prototypes, enabling simultaneous clustering and low-dimensional embedding of high-dimensional molecular representations [85]. • Captures non-linear structure via neighborhood-constrained competitive learning and supports map-based interpretation for chemotype exploration [85]. • Applied in chemoinformatics for organizing compound libraries and property landscapes [168] | • Requires selecting map size and training schedules (learning rate, neighborhood radius), since outcomes can be sensitive to initialization and hyperparameters [85]. • Topology preservation is not guaranteed, as map folding distortions can occur, especially under suboptimal training or complex manifolds [169]. • Training cost grows with the number of map units and feature dimension. Very large maps can become computationally heavy without approximate BMU search or parallelization [85] | $O(Tnmd)$ [85] | $O(md + n)$ [85] |
| | K-Medoids [60] | • More robust to outliers than K-Means, as it uses actual data points (medoids) as cluster centers [140]. • Provides better interpretability, since medoids represent real compounds within each cluster [141]. • Useful in chemical space analysis for identifying central, representative molecules in each group [170] | • Computationally expensive due to the need to calculate pairwise distances between all data points [171]. • Requires predefining the number of clusters, which can lead to poor performance if inaccurately set [172]. • Less effective in clustering non-globular or irregularly shaped datasets, despite its robustness to noise [140] | $O(dn^2 + Tkn^2)$ [173] | $O(n^2)$ [173] |
| | BIRCHBitBIRCH [9, 50] | • Uses tree structures for efficient summarization of large datasets [143]. • Memory-efficient and capable of handling massive chemical repositories (e.g., PubChem) [143, 174]. • Supports incremental clustering and streaming data, suitable for dynamic chemical libraries [143] | • Performs best with globular or spherical clusters; struggles with irregular cluster shapes [143]. • Requires a predefined number of clusters, which can be problematic in exploratory analyses [175]. • Sensitive to initial CF-tree construction and parameter settings, which can impact clustering quality [176] | $O(nd)$ [143] | $O(fd)$ [143] |

**Table 4** (continued)

| Category | Clustering technique | Strengths | Limitations | Time Complexity[1 2] | Space Complexity[1 2] |
|---|---|---|---|---|---|
| | Taylor-Butina [9, 144] | • Uses Tanimoto-based similarity on molecular fingerprints for structure-based compound grouping [137]. • Efficient for large molecular datasets, particularly in chemical space visualization and drug discovery [145]. • Effectively identifies chemical analogs and diverse molecular groups using structural features [177]. • Performs well with large-scale databases by pinpointing relevant molecule clusters for virtual screening [9] | • Less effective for irregular, non-globular cluster shapes or complex chemical structures [178]. • May produce less meaningful groupings when dealing with non-traditional molecular forms [178] | $O(dn^2 + n \log n)$ [144] | $O(n^2)$ [144] |
| | Bemis-Murcko Scaffold [3, 37, 66] | • Efficient data summarization [148]. • Scalable for million- to billion-scale datasets [3, 37]. • Enables scaffold diversity exploration [3, 37]. • Deterministic scaffold-based partitioning independent of descriptor or latent space | • Oversimplifies molecules by removing substituents and stereochemistry [3, 37]. • Can split related analogs into separate scaffolds [149]. • Limited for detailed chemical reaction information or conformational analysis [37, 149]. • Discards substituent-level detail, so molecules with identical core topology but different peripheral chemistry are grouped together | $O(nS)$ [179] | $O(n + g)$ [179] |
| Soft | Fuzzy (FCM) [33, 153] | • Allows data points to belong to multiple clusters with varying degrees of membership, useful for capturing partial similarities [152]. • Ideal for datasets where compounds share structural features across multiple chemical groups [153]. • Provides overlapping cluster boundaries, enabling more nuanced visualization of chemical space and molecular families [153] | • Computationally more intensive than hard clustering methods like K-Means due to iterative membership assignment [180]. • Performance is highly sensitive to the selection of the number of clusters and the membership function [181]. • Resource demands increase significantly when the method is applied to large chemical libraries and high-dimensional data [180] | $O(Tndk)$ [154, 182] | $O(nk + kd)$ [154] |
| | Responsibility Vectors-based Clustering (GTM-like) [46, 56, 57] | • Assigns soft cluster memberships through probability-based responsibility vectors [155]. • Well-suited for complex molecular datasets, such as DELs [57]. • Offers high flexibility by adapting to various chemical information structures and managing uncertainty [156] | • Computationally intensive due to the need to evaluate likelihoods across multiple clusters for every compound [183]. • Sensitive to data quality; noisy or poorly represented molecular features can lead to inaccurate clustering [184] | $O(TnLd)$ | $O(nL)$ |

**Table 4** (continued)

| Category | Clustering technique | Strengths | Limitations | Time Complexity[1,2] | Space Complexity[1,2] |
|---|---|---|---|---|---|
| Hierarchical | Agglomerative Hierarchical Clustering [32, 55, 59, 62, 64, 65] | • Begins with no predefined number of clusters, allowing flexible early-stage data exploration [157]. • Helps identify natural cluster groupings when the appropriate number of clusters is not obvious [160]. • Visualizes clustering structure through dendrograms, offering interpretability and insights into molecular similarities [159]. • Commonly used in cheminformatics and related fields for its interpretability and data structure exploration capabilities [161] | • Computationally expensive due to the need to calculate pairwise distances between all data points [157].• Sensitive to noise and outliers, which can distort dendrogram structure and result in misleading clusters [185]. • Challenging to interpret when applied to datasets with high variability or complex structure [186] | $O(dn^2 + n^3)$; $O(dn^2 + n^2 \log n)$ for the optimized implementation. [187] | $O(n^2)$ [187] |

[1] Definitions: $n$: Number of data points

$d$: Number of features

$k$: Number of clusters

$T$: Number of iterations

$m$: Number of map units (neurons)

$f$: Number of leaf entries

$S$: Cost to compute a scaffold

$g$: Number of unique scaffolds

$L$: Number of latent grid points

[2] Complexity not explicitly reported in the cited source was derived from our analysis of the algorithm as described in the referenced paper

an $O(dn^2)$ distance-computation phase), and is therefore rarely used beyond moderate $n$ without approximation or sampling [173].

To overcome these scaling constraints, BIRCH-style approaches avoid explicit all-pairs storage by streaming the data into a compact summary structure (the CF-tree), yielding near-linear construction time (often expressed as $O(nd)$ for fixed tree parameters) and overhead proportional to the number of retained leaf entries rather than $n^2$ [143]. BitBIRCH is a BIRCH-inspired algorithm that combines a CF-tree with the instant similarity (iSIM) mechanism to cluster binary fingerprints using Tanimoto similarity with near-linear scaling and a substantially smaller memory footprint. In benchmarks, BitBIRCH was more than three orders of magnitude faster than the RDKit implementation of Taylor-Butina on a library of around 1.5 million molecules, and an approximation variant was demonstrated to cluster on the order of one billion molecules in under five hours on modern hardware without degrading cluster quality [9]. Bemis-Murcko scaffold grouping provides an extreme-scale alternative by reducing each molecule to its core framework and performing deterministic scaffold-based partitioning, enabling efficient structural classification and diversity analysis across ultra-large datasets while avoiding pairwise similarity computations [3, 37].

Beyond specialized billion-scale algorithms, general-purpose iterative methods can remain practical when their per-iteration costs are linear in $n$ and when $k$ (or other latent sizes) is controlled. K-means has average time complexity $O(Tnkd)$ and modest memory overhead beyond storing the input features, which makes it amenable to mini-batch, distributed, and GPU-accelerated implementations for large chemical libraries. SOMs provide a related prototype-based alternative that also scales linearly in $n$ for a fixed map size, as the per-epoch training cost $O(nmd)$ is dominated by best-matching-unit (BMU) search and neighborhood updates [85].

Soft clustering methods introduce additional memory overhead: fuzzy c-means (FCM) typically requires storing an $n \times k$ membership matrix (overhead $O(nk)$) and has per-iteration cost on the order of $O(Tnkd)$, while responsibility-vector-based schemes (e.g., GTM-style responsibilities) store an $n \times L$ responsibility matrix (overhead $O(nL)$) with typical per-iteration cost $O(TnLd)$. Consequently, such soft methods can scale well in $n$ for moderate $k$ or $L$, but become memory-bound as the number of clusters and latent nodes grows.

In practice, large-scale clustering often combines algorithmic choices with implementation-level optimizations. To reduce per-comparison cost and memory, long fingerprints may be folded to shorter bit vectors (reducing memory at the cost of potential bit collisions), and fast bit operations (e.g., popcount-based Tanimoto) are used to accelerate similarity evaluation. To reduce the total number of comparisons and wall-clock time, batch, streaming, and distributed strategies split very large libraries into manageable shards that can be processed in parallel and then merged or reconciled via representative-based postprocessing [9, 27]. Finally, although hierarchical clustering remains valuable for interpretability, it is most realistically applied at scale to subsamples or to reduced representations (e.g., CF-tree leaf summaries, scaffold representatives, or cluster centroids), rather than directly to the full billion-molecule library due to its $O(n^2)$ memory requirement [143].

#### Evaluation metrics used to assess clustering quality

Evaluating clustering quality is essential for interpreting chemical space organization. Figure 9 summarizes the common evaluation metrics used to assess clustering quality, which fall into structural, statistical, and projection-fidelity categories. Molecular structural similarity metrics such as Tanimoto similarity [188] and Maximum Common Substructure (MCS) [189] are widely adopted in many studies to assess structural similarity and diversity of a dataset or grouping of molecules. Additionally, scaffold analysis provides a categorical grouping based on shared molecular frameworks rather than a continuous measure of pairwise similarity [2, 3, 18, 20, 24, 25, 27, 29, 30, 32, 37, 41, 44, 46, 47, 56–58, 66, 68].

Non-domain-specific statistical measures like ARI, Davies-Bouldin, Calinski-Harabasz, and Silhouette scores evaluate compactness and separation of molecular clusters [9, 40, 50, 55, 59, 62, 65]. Projection-based metrics, including Trustworthiness, Continuity, LCMC, and neighborhood-preservation indices (e.g., PNN, QNN), assess how well low-dimensional projections preserve molecular neighborhood relationships [8, 45]. Collectively, these metrics offer complementary understanding of cluster coherence, separability, and topological stability in chemical space exploration. In this context, these findings complete the response to **RQ4** by illustrating how clustering quality has been assessed across the reviewed studies.

## Discussion and recommendations

This section outlines the key findings from the reviewed studies, emphasizing dataset-size-dependent patterns in chemical space visualization pipelines. The analysis also outlines persistent challenges and future research directions throughout the field.

#### Implications of dataset scale

In the 56 reviewed studies, molecular fingerprints remain the dominant representation choice, reflecting their

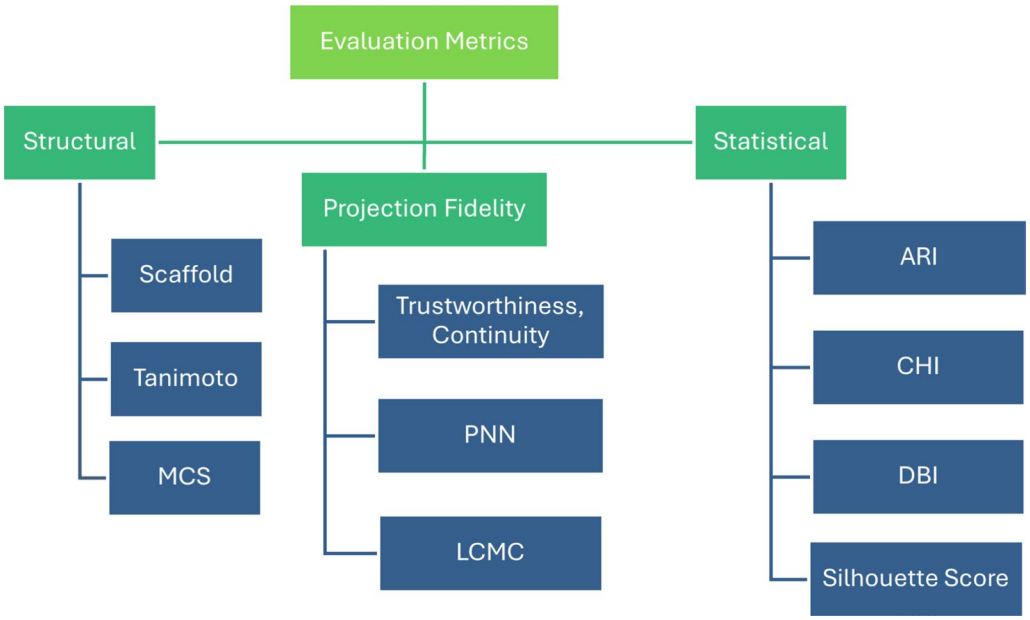



**Fig. 9** Taxonomy of evaluation metrics used to assess clustering quality

favorable trade-off between computational efficiency and chemical relevance, particularly for large libraries where similarity search and clustering must be executed at scale. Over time, the representation landscape shows a clear shift: early studies relied more heavily on handcrafted descriptors and domain-specific representations, whereas the 2011–2020 period marks a strong consolidation around fingerprint-based pipelines, consistent with the growing availability of million-scale public and vendor libraries. In the most recent period (2021–present), representation choices diversify again, with increased use of fragment descriptors, SMILES-based encodings, and learned embeddings, suggesting renewed emphasis on richer structure–property representations, while fingerprints continue to function as a scalable default baseline. Overall, the trend indicates that large dataset sizes push studies toward compact, index-based representations, while smaller datasets can accommodate more expressive and costlier featurizations.

DR shows a clear shift from scalable linear baselines toward neighborhood-preserving non-linear methods, constrained by computational and memory costs. PCA remains widely used across the reviewed studies, consistent with its efficient $O(nd)$ profile for fixed low-dimensional projections (i.e., small $c$) and its amenability to parallelization, making it practical for million-scale chemical libraries. As visualization goals increasingly prioritize local structure, t-SNE has gained prominence; however, its naive quadratic requirements (driven by pairwise similarities) and Barnes-Hut approximations that scale as $O(n \log n)$ per iteration limit its applicability to ultra-large datasets. More recently, UMAP has emerged as a more scalable non-linear alternative, with overall time $O(n \log n + Tnk)$ and working memory dominated by the sparse $k$-NN graph $O(nk)$, supporting its growing adoption in large-scale studies. At the extreme scale, TMAP reflects a shift toward graph-based layouts that enable visualization of ultra-large libraries where classical manifold learning becomes computationally prohibitive. These trends indicate a size-dependent trade-off: PCA persists as a robust baseline, t-SNE/UMAP are preferred when local neighborhood fidelity is prioritized and dataset size permits, and TMAP is utilized when extreme-scale visualization (multi-million and beyond) is required. It should be noted that DR techniques such as GTM can be applied to very large chemical libraries if they are implemented using hierarchical mapping strategies and precomputed map infrastructures, as has been demonstrated by recent work enabling navigation of chemical spaces that contain hundreds of millions of compounds [46].

Compared with featurization and DR, clustering shows the weakest convergence to a single dominant strategy: a small set of methods (hierarchical clustering, the K-means family, and SOM) appears repeatedly, but many studies use a wide range of other approaches. Early work relied mainly on SOM and K-means, while later studies show sustained use of hierarchical clustering together with a broader mix of methods, including scaffold-based clustering and other domain-driven variants, as well as

responsibility-vector-based strategies. This diversity is expected because clustering is often selected based on the analysis goal (e.g., scaffold coherence, neighborhood density, or activity-driven grouping) and is frequently used to compare alternative views of chemical space rather than as a single final answer. Dataset size further constrains feasible choices, as reflected by the computational and memory costs in our clustering Table 4: K-means scales as $O(Tnkd)$ time and $O(kd + n)$ space, whereas K-medoids scales as $O(dn^2 + Tkn^2)$ time and $O(n^2)$ space and becomes impractical for large libraries. For million-scale workflows, more scalable tree-based approaches such as BIRCH/BitBIRCH avoid explicit all-pairs storage and achieve near-linear construction time under fixed parameters, with memory proportional to the CF-tree size (number of retained leaf entries) rather than $n^2$. Taylor-Butina clustering can yield structurally coherent clusters with an appropriately tuned similarity threshold, but because it relies on pairwise fingerprint similarities and scales quadratically in time and space, it is rarely feasible at million-to-billion scale without approximations, batching, or specialized acceleration.

Taken together, the trends point to dataset size as the key constraint shaping practical combinations of featurization, DR, and clustering because it directly sets the feasible computation and memory budget. At small-to-medium scales, studies can afford richer representations (descriptors, fragments, hybrids, and in some cases learned embeddings) and more expensive methods (e.g., t-SNE, MDS variants, K-medoids, and some hierarchical variants) to improve interpretability and local structure discovery. As scale increases, practice converges toward representations and algorithms with sub-quadratic scaling and sparse non-$O(n^2)$ memory use: fingerprints (often compressed or folded), truncated PCA and increasingly UMAP and TMAP-style graph layouts, and clustering choices that avoid global all-pairs similarity. Consequently, pipelines favor centroid and prototype-based methods such as K-means and SOM, BIRCH-like summarization, and scaffold-based partitioning, or apply clustering to reduced representatives primarily to support visualization.

This review has several limitations that should be acknowledged. First, the analysis is based on the set of studies found through a structured but non-systematic search strategy, and so may not cover all relevant publications. Second, the trends in this work are based on the methodologies and dataset sizes used in the reviewed studies and not the full capabilities of the underlying algorithms. Third, the evaluation of DR and clustering techniques is dependent on their application in individual studies, which may differ with respect to their implementation, parameter settings, and data characteristics. Finally, rapidly evolving approaches, especially ones that rely on deep learning, might not be fully reflected in the existing literature. In spite of these limitations, the observed trends give a consistent basis for analyzing the computational trade-offs of different methods, which we formalize in the following paragraph.

To make these scale constraints explicit, we summarize time and space costs after substituting reference-extracted default parameters for each method, leaving only $n$ and $d$ (Tables 5 and 6). We rank algorithms by the resulting effective time complexity, while still reporting effective space because memory limits can be co-determining at large $n$. Although implementation details and hardware can shift constant factors, the resulting ordering clarifies which methods remain viable as $n$ grows. This synthesis motivates the next subsection: recommended chemical space visualization pipelines should be size-aware, and our proposed archetypes follow these complexity considerations. Notably, these recommendations reflect the methodological trends extracted from the reviewed studies and the standard algorithmic complexity of each method in its canonical form. In practice, scalable implementations that leverage distributed computing or GPU acceleration can extend many of these methods to larger datasets than their canonical complexity would suggest. However, such implementations typically introduce algorithmic modifications (e.g., hierarchical strategies, mini-batch processing, or approximate computations) that diverge from the canonical formulations on which our complexity analysis is based. A systematic comparison of accelerated implementations is beyond the scope of this review.

#### Pipeline archetypes by dataset scale

We propose a set of dataset-scale pipeline archetypes for chemical space visualization, guided by the usage trends observed across the 56 reviewed studies and the scaling constraints discussed in subsection Implications of dataset scale. Table 7 summarizes these archetypes as practical templates, listing common choices for molecular representation, DR, and clustering, together with brief notes on the dominant design objective in each dataset-size regime. These archetypes are not prescriptive but rather capture recurring combinations that balance interpretability, neighborhood fidelity, and computational feasibility as $n$ grows.

The archetypes are organized into five dataset-size regimes: small (< 5K), medium (5K–100K), large (100K–1 M), very large (1 M–1B), and ultra-large (> 1 B). For **small** datasets, studies can afford more expensive geometric objectives and interactive analysis, and therefore commonly pair chemically interpretable descriptors and fingerprints with DR methods that emphasize

**Table 5** Time and space complexities for DR algorithms after substituting default parameters extracted from references, leaving only $n$ and $d$

| Algorithm | Time (defaults substituted) | Space (defaults substituted) | Time rank |
|---|---|---|---|
| TMAP $k = 10, k_c = 10$ [190] | $O(n(100 + \log n))$ | $O(10n)$ | 1 |
| UMAP $k = 15, T = 200$ [191] | $O(n(3000 + \log n))$ | $O(15n)$ | 2 |
| PCA truncated $c = 2$ | $O(2nd)$ | $O(2n + 2d)$ | 3 |
| t-SNE Barnes-Hut $T = 1000, k = 90$ [192] | $O(n(d \log n + 1000 \log n))$ | $O(90n)$ | 4 |
| GTM $T = 100, g = 225$, $b = 100$ [8] | $O(22500nd + 100^4)$ | $O(225n + 100d)$ | 5 |
| VAE $T = 1000, z = 32$, $H = 50, L = 8$ [50] | $O(2 \cdot 10^7 n + 50000nd)$ | $O(50d + 21600)$ | 6 |
| pt-SNE $T = 80, B = 7000$, $H = 2048, L = 4$ [43] | $O(1.3 \cdot 10^9 n + 7.2 \cdot 10^5 nd)$ | $O(8 \cdot 10^7 + 7000d)$ | 7 |
| MDS | $O(n^3)$ | $O(n^2)$ | 8 |

We set $d \approx 2048$ for a common fingerprint length, since smaller or larger $d$ reorders PCA and UMAP. The complexity formulas can be found in Table 3

**Table 6** Time and space complexities for clustering algorithms after substituting default parameters extracted from references, leaving only $n$ and $d$

| Algorithm | Time (defaults substituted) | Space (defaults substituted) | Time rank |
|---|---|---|---|
| Bemis-Murcko $S = 30$ $g = 0.03n$ [3] | $O(30n)$ | $O(1.03n)$ | 1 |
| BitBIRCH $L =$ fraction of $n$ [9] | $O(nd)$ | $O(nd)$ | 2 |
| K-Means $T = 300, k = 8$ [40, 193] | $O(2400nd)$ | $O(8d + n)$ | 3 |
| Fuzzy (FCM) $T = 300, k = 8$ (same as K-means) | $O(2400nd)$ | $O(8d + 8n)$ | 3 |
| SOM $T = 10$ $m = 50 \times 50$ [194] | $O(25000nd)$ | $O(2500d + n)$ | 4 |
| RV GTM-like $T = 100$ $L = 24 \times 24$ [57] | $O(57600nd)$ | $O(576n)$ | 5 |
| Taylor-Butina | $O(dn^2 + n \log n)$ | $O(n^2)$ | 6 |
| HAC | $O(dn^2 + n^2 \log n)$ | $O(n^2)$ | 7 |
| K-Medoids $T = 300, k = 8$ (same as K-means) | $O(dn^2 + 2400n^2)$ | $O(n^2)$ | 8 |

The complexity formulas can be found in Table 4

faithful geometry or map-based interpretation (PCA, MDS, GTM, SOM, etc.). Clustering in this regime is correspondingly flexible, including fuzzy and hierarchical variants as well as similarity-threshold procedures (e.g., Taylor-Butina), reflecting an emphasis on interpretability and exploratory structure discovery.

For **medium** datasets, the archetype shifts toward neighborhood-preserving non-linear DR (UMAP and t-SNE) while retaining PCA as a baseline, and it introduces learned representations or latent models (e.g., VAE) in a subset of studies. Clustering remains diverse in this regime, with centroid and prototype-based methods (K-means, SOM) frequently used alongside fuzzy and hierarchical approaches. When higher-resolution structure is desired, quadratic or near-quadratic procedures (e.g., K-medoids, similarity-threshold clustering) may still be feasible.

For **large** datasets (100K–1 M), scalability becomes a primary constraint, and pipelines increasingly adopt graph- and index-based visualization (TMAP) and efficient baselines (truncated PCA, UMAP), with t-SNE constrained to subsamples when used. Clustering choices likewise prioritize near-linear memory and compute: scaffold-based partitioning (Bemis-Murcko) and BIRCH-style summarization (BitBIRCH) are common, with K-means used when an explicit partition is needed. More expensive clustering procedures (e.g., HAC or Taylor-Butina) are constrained to representatives rather than

**Table 7** Pipeline archetypes by dataset size

| Dataset Size | Featurization | DR | Clustering | Notes |
|---|---|---|---|---|
| Small (<5K) | Molecular descriptors (physicochemical, MOE), fingerprints (MACCS, ECFP4) | PCA, MDS, GTM, SOM (map-based visualization) | Fuzzy, SOM, Taylor-Butina, RV GTM-like, HAC | Emphasis on interpretability |
| Medium (5K–100K) | ECFP/Morgan, ISIDA fragments, MACCS, MQN fingerprints | UMAP, PCA, t-SNE, GTM, VAE | K-means, Fuzzy, SOM, Taylor-Butina, RV GTM-like, HAC, K-medoids | Shift toward nonlinear DR |
| Large (100K–1M) | ECFP4/6, ISIDA fragments, SMILES-based embeddings, RDKit descriptors | TMAP, PCA (truncated), UMAP, t-SNE on subsamples | Bemis-Murcko, BitBIRCH, K-means, HAC or Taylor-Butina on representatives | Balance neighborhood fidelity with near-linear memory and compute |
| Very Large (1M–1B) | MinHash (MHFP6), Folded fingerprints (Morgan/ECFP), compact descriptors | TMAP, PCA (truncated), UMAP on subsamples, **pretrained** projection models (t-SNE, VAE, pt-SNE) | Bemis-Murcko, BitBIRCH, K-means, HAC or Taylor-Butina on representatives | Focus on scalability; use approximations of quadratic all-pairs procedures |
| Ultra-Large (>1B) | MQN, SMIfp, fragment fingerprints, latent embeddings | TMAP, PCA (truncated), **pretrained** projection models (t-SNE, VAE, pt-SNE) | Bemis-Murcko, BitBIRCH | Avoid full all-pairs techniques; exploit parallelizable, near-linear techniques |

applied to all molecules. By representatives, we mean applying a scalable clustering procedure first, then selecting a subset of representative molecules (e.g., CF-tree summaries, scaffold representatives, or centroids).

For **very large** libraries (1 M–1B), the archetype is explicitly scalability-driven: representations favor compact or index-friendly fingerprints (e.g., folded fingerprints or MinHash-style schemes), and DR favors TMAP and truncated PCA, with UMAP applied to subsamples when needed for local neighborhood fidelity. When neural or parametric DR appears at this scale, it is most naturally used as a pretrained projection model whereby it is trained on a manageable subset and then used for scalable out-of-sample inference. Clustering at this scale prioritizes scaffold-based partitioning and summarization (Bemis-Murcko, BitBIRCH), with any quadratic procedures restricted to representative sets.

Finally, for **ultra-large** datasets (> 1B), pipelines avoid full all-pairs operations and rely on highly scalable representations (compact fingerprints, fragments, or latent embeddings), combined with TMAP, truncated PCA, and pretrained projections for out-of-sample mapping. Clustering is typically limited to extreme-scale partitioning and summarization (Bemis-Murcko, BitBIRCH), reflecting the practical need to keep both compute and memory sub-quadratic in $n$.

#### Gaps and future research directions

Despite substantial progress in DR, clustering, and large-scale chemical space mapping, several gaps remain that limit the reliability, coverage, and practical utility of chemical space visualizations. Across the reviewed literature, these gaps are tied to five recurring issues: how visualizations are evaluated, how molecules are represented, how methods scale to very large datasets, how reproducible analyses are, and how well current benchmarks cover diverse chemical domains. By examining the gaps discussed below and outlining future research directions aimed at improving evaluation consistency, representation richness, scalability, reproducibility, and dataset diversity in chemical space visualization, this subsection answers the final research question **RQ6**.

A major gap is the lack of consistent, structure-aware quantitative evaluation. Current practices frequently prioritize visual inspection and generic embedding metrics, but these approaches often lack a systematic assessment of how well low-dimensional proximity preserves intrinsic chemical relationships, such as scaffold hierarchies or Tanimoto similarity. This creates a risk of over-interpreting visually separated regions as chemically meaningful. Future work should more consistently report structure-based measures alongside visualization outputs, using chemical similarity as the reference notion of neighborhood. Examples include neighborhood-preservation metrics derived from a co-ranking matrix (e.g., PNN, QNN, LCMC), which compare $k$-NN rank structure in the embedding against $k$-NN rank structure under a reference distance in the original representation space [8]. These neighborhood-agreement scores can be made structure-aware by defining the reference neighborhoods using fingerprint Tanimoto similarity or scaffold-based similarity, rather than a generic metric in descriptor space. For clustering, complementary structure-aware diagnostics include within-cluster Tanimoto distributions, scaffold diversity, and enrichment of known chemical series within clusters or map regions. Adopting a standard set of such structure-aware metrics would reduce reliance on visual inspection, improve comparability across studies and datasets, and support more reliable benchmarking of both DR and clustering pipelines [38].

A second gap is the representation trade-off: fingerprints are efficient and scalable but can miss stereochemistry and 3D effects, while learned embeddings can be richer yet harder to interpret and sometimes computationally heavier to train and validate. Future work should benchmark hybrid representations that combine complementary signals (fingerprints, fragments, physicochemical descriptors, and learned embeddings) and clarify when the added complexity measurably improves neighborhood preservation, cluster coherence, or chemical series separation.

At million-to-billion scale, many DR and clustering methods become impractical without approximation or specialized algorithms. Priorities for future work include scalable neighborhood-based DR (e.g., approximate UMAP), clustering that avoids dense pairwise similarity matrices (e.g., BIRCH/BitBIRCH-style summarization), and workflows that favor coarse projections or graph layouts over exhaustive clustering at ultra-large scale. There is also an opportunity to develop more chemical-domain-specific DR and clustering methods that exploit scaffold structure or fingerprint similarity while remaining scalable and interpretable.

Reproducibility and transparency also remain inconsistent. Hyperparameters (e.g., DR settings, distance metrics, fingerprint length, sampling strategies), preprocessing steps, and filtering choices can strongly affect visual outcomes, yet reporting varies widely and code availability is mixed. Future work should adopt clearer reporting standards for chemical space visualization studies, including complete specification of representations, preprocessing, hyperparameters, and evaluation criteria, and should prioritize open-source implementations and reusable benchmark datasets so that results can be replicated and compared.

Finally, there are opportunities to broaden chemical space coverage beyond common drug-like and commercially curated libraries. Many visualizations focus on bioactivity-rich datasets and vendor-accessible collections, while natural products, metal-containing compounds, environmental chemicals, and other underrepresented domains receive less attention. Expanding benchmark datasets and validation tasks to better cover these regions would improve generality and reduce domain bias in both representations and visualization methods.

## Conclusion

Chemical space visualization has moved from small, interactive studies to workflows that must handle million- to multi-billion-compound libraries under tight compute and memory limits. By reviewing 56 studies published between 2000 and 2025, we treat chemical space visualization as an end-to-end pipeline that includes dataset selection, featurization, dimensionality reduction (DR), clustering, and evaluation. Several consistent patterns emerge. Fingerprints remain the most common representation at scale, while fragments, SMILES-derived encodings, and learned embeddings are increasingly used when richer structure–property signals are needed and the added cost is justified. DR has expanded beyond PCA baselines toward neighborhood-preserving methods such as t-SNE and UMAP, and toward extreme-scale layout approaches such as TMAP when dense pairwise computations become too expensive. Clustering remains diverse and strongly driven by the analysis goal: hierarchical clustering, K-means, and SOM appear frequently, while large-library settings increasingly motivate scalable alternatives such as BIRCH/BitBIRCH and scaffold-based partitioning that avoid explicit $O(n^2)$ similarity construction.

A central contribution of this review is the set of dataset-scale pipeline archetypes distilled from these observed patterns. Rather than presenting methods in isolation, the archetypes summarize compatible choices for representation, DR, and clustering across practical size regimes as scale increases, and clarify the key trade-offs that constrain method selection. These templates are intended to support new studies by indicating when simple, interpretable baselines are sufficient, when neighborhood-preserving DR is worth the added cost, and when ultra-large settings require layouts and grouping strategies that prioritize scalable organization over exhaustive partitioning.

This review also highlights gaps that limit how comparable and reliable chemical space visualizations are across studies. Many papers still rely heavily on visual inspection without consistent structure-aware quantitative checks. Reporting of preprocessing steps and key hyperparameters varies widely, even though these choices can change the map and the resulting groupings substantially. Benchmarking is also uneven across chemical domains, with many studies concentrating on drug-like and commercially curated libraries. Progress will require clearer reporting standards, more diverse benchmark datasets and tasks, and scalable methods and evaluation protocols that remain chemically meaningful and interpretable.

**Abbreviations**

| | |
|---|---|
| ARI | Adjusted Rand Index |
| BCUT | Burden–Eigenvalue/Charge Tensor |
| BRICS | Breaking Retrosynthetically Interesting Chemical Substructures |
| CF | Clustering Feature |
| CHI | Calinski-Harabasz Index |
| cl-MDS | Cluster-Based Multidimensional Scaling |
| DBI | Davies-Bouldin Index |
| DEL | DNA-Encoded Library |
| DR | Dimensionality Reduction |
| ECFP | Extended Connectivity Fingerprint |
| GMM | Gaussian Mixture Model |
| GTM | Generative Topographic Mapping |
| HGTM | Hierarchical Generative Topographic Mapping |
| KDE | Kernel Density Estimation |
| LCMC | Local Continuity Meta-Criterion |
| MACCS | Molecular ACCess System keys |
| MAE | Mean Absolute Error |
| MCS | Maximum Common Substructure |
| MDS | Multidimensional Scaling |
| MOE | Molecular Operating Environment |
| MQN | Molecular Quantum Numbers |
| MST | Minimum Spanning Tree |
| NAMS | Non-contiguous Atom Matching Structural Similarity |
| NCI | National Cancer Institute |
| PCA | Principal Component Analysis |
| PCVG | Principal Component Variable Grouping |
| PNN | Neighborhood Preservation Score |
| pt-SNE | Parametric t-Distributed Stochastic Neighbor Embedding |
| QNN | Quantized Neighborhood Preservation |
| RMSD | Root Mean Square Deviation |
| ROC | Receiver Operating Characteristic curve |
| RXNFP | Reaction Fingerprint |
| SA | Synthetic Accessibility |
| SALI | Structure-Activity Landscape Index |
| SMILES | Simplified Molecular Input Line Entry System |
| SOAP | Smooth Overlap of Atomic Positions |
| SOM | Self-Organizing Map |
| TMAP | Tree Map Projection |
| t-SNE | T-Distributed Stochastic Neighbor Embedding |
| UMAP | Uniform Manifold Approximation and Projection |
| VAE | Variational Autoencoder |
| WDI | World Drug Index |

**Author contributions**

M.AS. prepared the manuscript text and analysis. M.AN. reviewed and edited the manuscript. M.AH. reviewed and edited the manuscript.

**Funding**

Open Access funding provided by the King Fahd University of Petroleum and Minerals' Deanship of Research.

**Data availability**

No datasets were generated or analysed during the current study.

## Declarations

**Competing of interests**
The authors declare no conflict of interest.

Received: 3 January 2026 Accepted: 6 May 2026
Published online: 02 June 2026

**References**

1. Medina-Franco J, Martinez-Mayorga K, Giulianotti M, Houghten R, Pinilla C (2008) Visualization of the chemical space in drug discovery. Curr Comput Aided Drug Design 4:322–333. https://doi.org/10.2174/157340908786786010
2. Reymond JL (2015) The chemical space project. Acc Chem Res 48:722–730. https://doi.org/10.1021/ar500432k
3. Tingle BI, Tang KG, Castanon M, Gutierrez JJ, Khurelbaatar M, Dandarchuluun C, Moroz YS, Irwin JJ (2023) Zinc-22-a free multi-billion-scale database of tangible compounds for ligand discovery. J Chem Inf Model 63:1166–1176. https://doi.org/10.1021/acs.jcim.2c01253
4. Weininger D (1988) Smiles, a chemical language and information system. 1. introduction to methodology and encoding rules. J Chem Inf Comput Sci 28(1):31–36. https://doi.org/10.1021/ci00057a005
5. Wojtuch A, Danel T, Podlewska S (2023) Extended study on atomic featurization in graph neural networks for molecular property prediction. J Cheminform 15:1–27. https://doi.org/10.1186/S13321-023-00751-7/TABLES/15
6. Maaten LVD, Hinton G (2008) Visualizing data using t-sne. J Mach Learn Res 9:2579–2605
7. McInnes, L., Healy, J., Melville, J.: Umap: Uniform manifold approximation and projection for dimension reduction. ArXiv e-prints (2018) arXiv:1802.03426 [stat.ML]
8. Orlov AA, Akhmetshin TN, Horvath D, Marcou G, Varnek A (2025) From high dimensions to human insight: exploring dimensionality reduction for chemical space visualization. Mol Inf 44(1):202400265. https://doi.org/10.1002/minf.202400265
9. Pérez KL, Jung V, Chen L, Huddleston K, Miranda-Quintana RA (2025) Bitbirch: efficient clustering of large molecular libraries. Digital Discovery. https://doi.org/10.1039/D5DD00030K
10. Rousseeuw PJ (1987) Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. J Comput Appl Math 20:53–65. https://doi.org/10.1016/0377-0427(87)90125-7
11. Daszykowski M, Walczak B, Massart DL (2003) Projection methods in chemistry. Chemom Intell Lab Syst 65:97–112. https://doi.org/10.1016/S0169-7439(02)00107-7
12. Ivanenkov YA, Savchuk NP, Ekins S, Balakin KV (2009) Computational mapping tools for drug discovery. Drug Discovery Today 14:767–775. https://doi.org/10.1016/j.drudis.2009.05.016
13. Osolodkin DI, Radchenko EV, Orlov AA, Voronkov AE, Palyulin VA, Zefirov NS (2015) Progress in visual representations of chemical space. Expert Opin Drug Discov 10:959–973. https://doi.org/10.1517/17460441.2015.1060216
14. Lachance H, Wetzel S, Kumar K, Waldmann H (2012) Charting, navigating, and populating natural product chemical space for drug discovery. J Med Chem 55:5989–6001. https://doi.org/10.1021/jm300288g
15. Medina-Franco JL, Sánchez-Cruz N, López-López E, Díaz-Eufracio BI (2022) Progress on open chemoinformatic tools for expanding and exploring the chemical space. J Comput Aided Mol Des 36:341–354. https://doi.org/10.1007/s10822-021-00399-1
16. Saldívar-González FI, Medina-Franco JL (2022) Approaches for enhancing the analysis of chemical space for drug discovery. Expert Opin Drug Discov 17(7):789–798. https://doi.org/10.1080/17460441.2022.2084608
17. Sosnin S (2025) Chemical space visual navigation in the era of deep learning and big data. Drug Discovery Today 30(7):104392. https://doi.org/10.1016/j.drudis.2025.104392
18. Agrafiotis DK, Rassokhin DN, Lobanov VS (2001) Multidimensional scaling and visualization of large molecular similarity tables. J Comput Chem 22(5):488–500. https://doi.org/10.1002/1096-987X(20010415)22:5<488::AID-JCC1020>3.0.CO;2-4
19. Oprea TI, Gottfries J (2001) Chemography: the art of navigating in chemical space. J Comb Chem 3:157–166. https://doi.org/10.1021/cc0000388
20. Givehchi A, Dietrich A, Wrede P, Schneider G (2003) Chemspaceshuttle: a tool for data mining in drug discovery by classification, projection, and 3d visualization. QSAR Comb Sci 22:549–559. https://doi.org/10.1002/qsar.200310007
21. Harper G, Bravi GS, Pickett SD, Hussain J, Green DVS (2004) The reduced graph descriptor in virtual screening and data-driven clustering of high-throughput screening data. J Chem Inf Comput Sci 44:2145–2156. https://doi.org/10.1021/ci049860f
22. Maniyar DM, Nabney IT, Williams BS, Sewing A (2006) Data visualization during the early stages of drug discovery. J Chem Inf Model 46:1806–1818. https://doi.org/10.1021/ci050471a
23. Ivosev G, Burton L, Bonner R (2008) Dimensionality reduction and visualization in principal component analysis. Anal Chem 80:4933–4944. https://doi.org/10.1021/ac800110w
24. Singh N, Guha R, Giulianotti MA, Pinilla C, Houghten RA, Medina-Franco JL (2009) Chemoinformatic analysis of combinatorial libraries, drugs, natural products, and molecular libraries small molecule repository. J Chem Inf Model 49:1010–1024. https://doi.org/10.1021/ci800426u
25. Virshup AM, Contreras-García J, Wipf P, Yang W, Beratan DN (2013) Stochastic voyages into uncharted chemical space produce a representative library of all possible drug-like compounds. J Am Chem Soc 135:7296–7303. https://doi.org/10.1021/ja401184g
26. Malhat MG, Mousa HM, El-Sisi AB (2014) Clustering of chemical data sets for drug discovery. In: 2014 9th international conference on informatics and systems, p. 11. https://doi.org/10.1109/INFOS.2014.7036702. IEEE
27. Gaspar HA, Baskin II, Marcou G, Horvath D, Varnek A (2015) Chemical data visualization and analysis with incremental generative topographic mapping: Big data challenge. J Chem Inf Model 55:84–94. https://doi.org/10.1021/ci500575y
28. De S, Bartók AP, Csányi G, Ceriotti M (2016) Comparing molecules and solids across structural and alchemical space. Phys Chem Chem Phys 18:13754–13769. https://doi.org/10.1039/c6cp00415f
29. Naveja JJ, Medina-Franco JL (2017) Chemmaps: towards an approach for visualizing the chemical space based on adaptive satellite compounds. F1000Research 6, 1134. https://doi.org/10.12688/f1000research.12095.1
30. Awale M, Probst D, Reymond JL (2017) Webmolcs: a web-based interface for visualizing molecules in three-dimensional chemical spaces. J Chem Inf Model 57:643–649. https://doi.org/10.1021/acs.jcim.6b00690
31. Smith JS, Nebgen B, Lubbers N, Isayev O, Roitberg AE (2018) Less is more: Sampling chemical space with active learning. Journal of Chemical Physics 148. https://doi.org/10.1063/1.5023802
32. Saldívar-González FI, Valli M, Andricopulo AD, Bolzani VDS, Medina-Franco JL (2019) Chemical space and diversity of the nubbe database: a chemoinformatic characterization. J Chem Inf Model 59:74–85. https://doi.org/10.1021/acs.jcim.8b00619
33. Bandyopadhyay D, Kreatsoulas C, Brady PG, Boyer J, He Z, Scavello G, Peryea T, Jadhav A, Nguyen DT, Guha R (2019) Scaffold-based analytics: Enabling hit-to-lead decisions by visualizing chemical series linked across large datasets. J Chem Inf Model 59:4880–4892. https://doi.org/10.1021/acs.jcim.9b00243
34. Kausar S, Falcao AO (2019) A visual approach for analysis and inference of molecular activity spaces. J Cheminform 11:1–17. https://doi.org/10.1186/S13321-019-0386-Z/FIGURES/6
35. Karlov D, Sosnin S, Tetko I, Fedorov M (2019) Chemical space exploration guided by deep neural networks. RSC Adv 9:5151–5157. https://doi.org/10.1039/c8ra10182e
36. Pracht P, Bohle F, Grimme S (2020) Automated exploration of the low-energy chemical space with fast quantum chemical methods. Phys Chem Chem Phys 22:7169–7192. https://doi.org/10.1039/c9cp06869d
37. Irwin JJ, Tang KG, Young J, Dandarchuluun C, Wong BR, Khurelbaatar M, Moroz YS, Mayfield J, Sayle RA (2020) Zinc20-a free ultralarge-scale chemical database for ligand discovery. J Chem Inf Model 60(12):6065–6073. https://doi.org/10.1021/acs.jcim.0c00675. (**PMID: 33118813**)

38. Probst D, Reymond J-L (2020) Visualization of very large high-dimensional data sets as minimum spanning trees. J Cheminform 12(1):12. https://doi.org/10.1186/s13321-020-0416-x
39. Fraux G, Cersonsky R, Ceriotti M (2020) Chemiscope: interactive structure-property explorer for materials and molecules. J Open Sourc Softw 5:2117. https://doi.org/10.21105/joss.02117
40. Nuñez JR, McGrady M, Yesiltepe Y, Renslow RS, Metz TO (2020) Chespa: streamlining expansive chemical space evaluation of molecular sets. J Chem Inf Model 60:6251–6257. https://doi.org/10.1021/acs.jcim.0c00899
41. Shrivastava AD, Kell DB (2021) Fragnet, a contrastive learning-based transformer model for clustering, interpreting, visualizing, and navigating chemical space. Molecules. https://doi.org/10.3390/molecules26072065
42. Schwaller P, Probst D, Vaucher AC, Nair VH, Kreutter D, Laino T, Reymond JL (2021) Mapping the space of chemical reactions using attention-based neural networks. Nat Mach Intell 3:144–152. https://doi.org/10.1038/s42256-020-00284-w
43. Andronov M, Fedorov M, Sosnin S (2021) Exploring chemical reaction space with reaction difference fingerprints and parametric t-sne. ACS Omega 6:30743–30751. https://doi.org/10.1021/acsomega.1c04778
44. Zabolotna Y, Lin A, Horvath D, Marcou G, Volochnyuk DM, Varnek A (2021) Chemography: searching for hidden treasures. J Chem Inf Model 61:179–188. https://doi.org/10.1021/acs.jcim.0c00936
45. Sabando MV, Ulbrich P, Selzer M, Byska J, Mican J, Ponzoni I, Soto AJ, Ganuza ML, Kozlikova B (2021) Chemva: interactive visual analysis of chemical compound similarity in virtual screening. IEEE Trans Visual Comput Graphics 27:891–901. https://doi.org/10.1109/TVCG.2020.3030438
46. Zabolotna Y, Bonachera F, Horvath D, Lin A, Marcou G, Klimchuk O, Varnek A (2022) Chemspace atlas: Multiscale chemography of ultralarge libraries for drug discovery. J Chem Inf Model 62:4537–4548. https://doi.org/10.1021/acs.jcim.2c00509
47. Wang Y, Xiong Y, Garcia EAL, Wang Y, Butch CJ (2022) Drug chemical space as a guide for new herbicide development: a cheminformatic analysis. J Agric Food Chem 70:9625–9636. https://doi.org/10.1021/acs.jafc.2c01425
48. Cihan Sorkun M, Mullaj D, Koelman JMVA, Er S (2022) Chemplot, a python library for chemical space visualization. Chem Methods 2(7), 202200005. https://doi.org/10.1002/cmtd.202200005
49. McGrady MY, Colby SM, Nuñez JR, Renslow RS, Metz TO (2022) AI for chemical space gap filling and novel compound generation. https://arxiv.org/abs/2201.12398
50. Hadipour H, Liu C, Davis R, Cardona ST, Hu P (2022) Deep clustering of small molecules at large-scale via variational autoencoder embedding and k-means. BMC Bioinform. https://doi.org/10.1186/s12859-022-04667-1
51. Scalfani VF, Patel VD, Fernandez AM (2022) Visualizing chemical space networks with rdkit and networkx. J Cheminform. https://doi.org/10.1186/s13321-022-00664-x
52. Ishfaq M, Halawa MI, Ahmad A, Rasool A, Manzoor R, Ullah K, Guan Y (2023) Generation of chemical space of compounds for prostate cancer treatment: biological activity prediction, clustering, and visualization of chemical space. ACS Omega 8:39408–39419. https://doi.org/10.1021/acsomega.3c05056
53. Gaytán-Hernández D, Chávez-Hernández AL, López-López E, Miranda-Salas J, Saldívar-González FI, Medina-Franco JL (2023) Art driven by visual representations of chemical space. J Cheminform. https://doi.org/10.1186/s13321-023-00770-4
54. Choubisa H, Abed J, Mendoza D, Matsumura H, Sugimura M, Yao Z, Wang Z, Sutherland BR, Aspuru-Guzik A, Sargent EH (2023) Accelerated chemical space search using a quantum-inspired cluster expansion approach. Matter 6:605–625. https://doi.org/10.1016/j.matt.2022.11.031
55. Domingo-Fernández D, Gadiya Y, Mubeen S, Healey D, Norman BH, Colluru V (2023) Exploring the known chemical space of the plant kingdom: insights into taxonomic patterns, knowledge gaps, and bioactive regions. J Cheminform. https://doi.org/10.1186/s13321-023-00778-w
56. Pikalyova R, Zabolotna Y, Horvath D, Marcou G, Varnek A (2023) Chemical library space: definition and dna-encoded library comparison study case. J Chem Inf Model 63:4042–4055. https://doi.org/10.1021/acs.jcim.3c00520
57. Pikalyova R, Zabolotna Y, Horvath D, Marcou G, Varnek A (2023) Metagtm: visualization and analysis of the chemical library space. J Chem Inf Model. https://doi.org/10.1021/acs.jcim.3c00719
58. Jalala VK, Muraleedharan K (2024) Unleashing the potential of cheminformatic analysis for mycobacterium tuberculosis inhibitors: insights into chemical space and structural diversity. Hybrid Adv 6:100235. https://doi.org/10.1016/j.hybadv.2024.100235
59. Villares M, Saunders CM, Fey N (2024) Comparison of dimensionality reduction techniques for the visualisation of chemical space in organometallic catalysis. Artific Intell Chem 2:100055. https://doi.org/10.1016/j.aichem.2024.100055
60. Hernández-León P, Caro MA (2024) Cluster-based multidimensional scaling embedding tool for data visualization. Phys Scr. https://doi.org/10.1088/1402-4896/ad432e
61. Wang X, Strobel M, Aron AT, Phelan VV, Acharya DD, Brown CJ, Clevenger K, Hu J, Kretsch A, Mahood EH, Menegatti C, Xiong Q, Wang M (2024) Network topology evaluation and transitive alignments for molecular networking. J Am Soc Mass Spectrom. https://doi.org/10.1021/jasms.4c00208
62. Mubashir T, Tahir MH, Shafiq Z, Dewidar AZ, El-ansary HO, Janjua MRSA (2024) An efficient framework to design near-ir monomers for polymer solar cells with the help of machine learning, virtual screening and chemical space visualization. J Photochem Photobiol, A. https://doi.org/10.1016/j.jphotochem.2023.115285
63. Pan W, Li Y (2024) Chemical space of small molecule exploration for the elucidation of cough suppressant pharmacology of ying su ke/zi. J Herbal Med. https://doi.org/10.1016/j.hermed.2023.100784
64. Manen-Freixa L, Moliner-Cubel S, Gamo FJ, Crespo B, Borrell JI, Teixidó J, Estrada-Tejedor R (2024) Exploring the unexplored chemical space: rational identification of new tafenoquine analogs with antimalarial properties. Bioorg Chem. https://doi.org/10.1016/j.bioorg.2024.107472
65. Saqib M, Sagir M, Joshi ML, Bashir S, Halawa MI, Ali S, Elansary HO, Kamal GM (2024) Screening and designing of a large chemical space of organic semiconductors for photodetectors. Mater Today Commun. https://doi.org/10.1016/j.mtcomm.2024.108062
66. Aouidate A (2024) Exploring the chemical space of braf inhibitors: a cheminformatic and machine learning analysis. J Mol Liq. https://doi.org/10.1016/j.molliq.2024.124705
67. Zahoránszky-Kőhalmi G, Wan KK, Godfrey AG (2024) Hilbert-curve assisted structure embedding method. J Cheminform. https://doi.org/10.1186/s13321-024-00850-z
68. Mahmood A, Naeem S, Javed A, Shafiq Z, El-Sheikh MA, Elansary HO, Janjua MRSA (2024) Chemical similarity-based design of materials for organic solar cells: visualizing the generated chemical space of polymers. Mater Today Commun. https://doi.org/10.1016/j.mtcomm.2024.108403
69. Gaulton A, Bellis LJ, Bento AP, Chambers J, Davies M, Hersey A, Light Y, McGlinchey S, Michalovich D, Al-Lazikani B, Overington JP (2011) Chembl: a large-scale bioactivity database for drug discovery. Nucleic Acids Res 40(D1):1100–1107. https://doi.org/10.1093/nar/gkr777
70. Kim S, Thiessen PA, Bolton EE, Chen J, Fu G, Gindulyte A, Han L, He J, He S, Shoemaker BA, Wang J, Yu B, Zhang J, Bryant SH (2015) Pubchem substance and compound databases. Nucleic Acids Res 44(D1):1202–1213. https://doi.org/10.1093/nar/gkv951
71. Wishart DS, Knox C, Guo AC, Shrivastava S, Hassanali M, Stothard P, Chang Z, Woolsey J (2006) Drugbank: a comprehensive resource for in silico drug discovery and exploration. Nucleic Acids Res 34:668–672. https://doi.org/10.1093/nar/gkj067
72. Irwin JJ, Shoichet BK (2005) Zinc - a free database of commercially available compounds for virtual screening. J Chem Inf Model 45(1):177–182. https://doi.org/10.1021/ci049714+. (**PMID: 15667143**)
73. Enamine: Enamine REAL Compounds. Accessed: 2026-03-20. https://enamine.net/compound-collections/real-compounds
74. Grygorenko OO, Radchenko DS, Dziuba I, Chuprina A, Gubina KE, Moroz YS (2020) Generating multibillion chemical space of readily accessible screening compounds. Iscience 23(11):101681
75. Landrum G, et al. (2025) RDKit: open-source cheminformatics. https://www.rdkit.org
76. Todeschini R, Consonni V (2009) Molecular descriptors for chemoinformatics. Wiley-VCH, Weinheim. https://doi.org/10.1002/9783527628766

77. Consonni V, Ballabio D, Todeschini R (2023) Chapter 12—chemical space and molecular descriptors for qsar studies. In: Roy, K. (ed.) Chem-informatics, QSAR and Machine Learning Applications for Novel Drug Development, pp. 303–327. Academic Press. https://doi.org/10.1016/B978-0-443-18638-7.00022-0 . https://www.sciencedirect.com/science/article/pii/B9780443186387000220
78. Rogers D, Hahn M (2010) Extended-connectivity fingerprints. J Chem Inf Model 50(5):742–754. https://doi.org/10.1021/ci100050t. (**PMID: 20426451**)
79. Lin B, Yan S, Zhen B (2025) A machine learning method for predicting molecular antimicrobial activity. Sci Rep 15(1):6559
80. Vogt M, Bajorath J (2007) Bayesian screening for active compounds in high-dimensional chemical spaces combining property descriptors and molecular fingerprints. Chem Biol Drug Design. https://doi.org/10.1111/j.1747-0285.2007.00602.x
81. Hu K, He Y, Wei J, Sun C, Geng J, Wei L, Su R (2025) Bfgtp: a bert-guided two-stage molecular representation learning framework for toxicity prediction. IEEE J Biomed Health Inform. https://doi.org/10.1109/JBHI.2025.3556766
82. Bair E, Hastie T, Paul D, Tibshirani R (2006) Prediction by supervised principal components. J Am Stat Assoc 101(473):119–137. https://doi.org/10.1198/016214505000000628
83. Cheng Y, Wang X, Xia Y (2021) Supervised t-distributed stochastic neighbor embedding for data visualization and classification. INFORMS J Comput 33(2):566–585
84. McInnes L, Healy J, Saul N, Großberger L (2018) Umap: uniform manifold approximation and projection. J Open Source Softw 3:861. https://doi.org/10.21105/JOSS.00861
85. Kohonen T (2001) Self-organizing maps, 3rd edn. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-56927-2
86. scikit-learn developers: manifold Learning methods on a severed sphere. https://scikit-learn.org/stable/auto_examples/manifold/plot_manifold_sphere.html
87. Gewers FL, Ferreira GR, Arruda HFD, Silva FN, Comin CH, Amancio DR, Costa LD (2018) Principal component analysis ACM Computing Surveys (CSUR) 54:1–34. https://doi.org/10.1145/3447755
88. Granato D, Santos JS, Escher GB, Ferreira BL, Maggio R (2018) Use of principal component analysis (pca) and hierarchical cluster analysis (hca) for multivariate association between bioactive compounds and functional properties in foods: a critical perspective. Trends Food Sci Technol 72:83–90. https://doi.org/10.1016/J.TIFS.2017.12.006
89. Kai-hong G (2007) Limitations of pca in solving the multiple correlations between variables. J Comput Appl
90. Harkat MF, Kouadri A, Fezai R, Mansouri M, Nounou H, Nounou M (2020) Machine learning-based reduced kernel pca model for nonlinear chemical process monitoring. J Control Autom Electr Syst 31:1196–1209. https://doi.org/10.1007/s40313-020-00604-w
91. Ahmad W (2020) Evaluating chemical descriptors with the loss-data framework
92. Yoo C, Shahlaei M (2018) The applications of pca in qsar studies: a case study on ccr5 antagonists. Chem Biol Drug Design 91:137–152. https://doi.org/10.1111/cbdd.13064
93. Olivon F, Elie N, Grelier G, Roussi F, Litaudon M, Touboul D (2018) Metgem software for the generation of molecular networks based on the t-sne algorithm. Anal Chem 90(23):13900–13908. https://doi.org/10.1021/acs.analchem.8b03099
94. Robinson I, Pierce-Hoffman E (2020) Tree-sne: hierarchical clustering and visualization using t-sne.ArXiv abs/2002.05687
95. Kimura M (2021) Generalized t-sne through the lens of information geometry. IEEE Access 9:129619–129625. https://doi.org/10.1109/ACCESS.2021.3113397
96. Cai T, Ma R (2021) Theoretical foundations of t-sne for visualizing high-dimensional clustered data. ArXiv **abs/2105.07536**
97. Renqiang Min M, Guo H, Shen D (2018) Parametric t-distributed stochastic exemplar-centered embedding. Joint European conference on machine learning and knowledge discovery in databases. Springer, pp 477–493
98. Garriga J, Ferre FB (2024) Towards a comprehensive visualisation of structure in large scale data sets. Mach Learn Sci Technol. https://doi.org/10.1088/2632-2153/ad6fea
99. Lai C-H, Kuo M, Lien Y-H, Su K-A, Wang Y-S (2022) Parametric dimension reduction by preserving local structure. 2022 IEEE visualization and visual analytics (VIS). pp 75–79. https://doi.org/10.1109/VIS54862.2022.00024
100. Becht E, Dutertre C, Kwok I, Ng L, Ginhoux F, Newell E (2018) Evaluation of umap as an alternative to t-sne for single-cell data. bioRxiv https://doi.org/10.1101/298430
101. Appleby G, Espadoto M, Chen R, Goree S, Telea A, Anderson EW, Chang R (2021) Hypernp: interactive visual exploration of multidimensional projection hyperparameters. Comput Graphics Forum. https://doi.org/10.1111/cgf.14531
102. Jackson A (2019) The mathematics of umap. Digital asset: https://adele-jacksoncom/files/Maths_of_UMAP.pdf (open in a new window)
103. Charalampidis D (2021) Visualizing population substructures using multidimensional scaling and data smoothing. 2021 international conference on computational science and computational intelligence (CSCI). pp 324–329. https://doi.org/10.1109/CSCI54926.2021.00127
104. Jakaitiene A, Sangiovanni M, Guarracino MR, Pardalos PM (2016) Multidimensional scaling for genomic data. In: Pardalos, P.M., Zhigljavsky, A., Zilinskas, J. (eds.) Advances in stochastic and deterministic global optimization. Springer optimization and its applications, pp. 129–139. Springer, Cham. https://doi.org/10.1007/978-3-319-29975-4
105. Krzanowski W (2006) Sensitivity in metric scaling and analysis of distance. Biometrics. https://doi.org/10.1111/j.1541-0420.2005.00398.x
106. Kireeva N, Baskin I, Baskin I, Gaspar HA, Horvath D, Marcou G, Varnek A (2012) Generative topographic mapping (gtm): universal tool for data visualization, structure-activity modeling and dataset comparison. Mol Inf. https://doi.org/10.1002/minf.201100163
107. Horvath D, Marcou G, Varnek A (2017)Generative topographic mapping approach to chemical space analysis, pp. 167–199. Springer, Cham.https://doi.org/10.1007/978-3-319-56850-8_6
108. Bishop CM, Svensén M, Williams CKI (1998) Developments of the generative topographic mapping. Neurocomputing 21(1):203–224. https://doi.org/10.1016/S0925-2312(98)00043-5
109. Wei R, Mahmood A (2021) Recent advances in variational autoencoders with representation learning for biomedical informatics: a survey. IEEE Access 9:4939–4956. https://doi.org/10.1109/ACCESS.2020.3048309
110. Ullanat V (2020) Variational autoencoder as a generative tool to produce de-novo lead compounds for biological targets. 14th international conference on innovations in information technology (IIT). pp 102–107. https://doi.org/10.1109/IIT50501.2020.9299078
111. Wu Z, Cao L, Qi L (2023) evae: evolutionary variational autoencoder. IEEE Trans Neural Netw Learn Syst 36:3288–3299. https://doi.org/10.1109/TNNLS.2024.3359275
112. Biswas A, Vasudevan R, Ziatdinov M, Kalinin S (2022) Optimizing training trajectories in variational autoencoders via latent Bayesian optimization approach. Mach Learn Sci Technol. https://doi.org/10.1088/2632-2153/acb316
113. Choi S, Lee C, Lee J-M, Park JH, Lee I-B (2005) Fault detection and identification of nonlinear processes based on kernel pca. Chemom Intell Lab Syst 75:55–67. https://doi.org/10.1016/J.CHEMOLAB.2004.05.001
114. Li X, Wang S, Cai Y (2019) Tutorial: complexity analysis of singular value decomposition and its variants.(arXiv preprint) arXiv:1906.12085
115. Candel G, Naccache D(2021) Index t-sne: tracking dynamics of high-dimensional datasets with coherent embeddings. ArXiv **abs/2109.10538**
116. Chan D, Rao R, Huang F, Canny J (2018) T-sne-cuda: Gpu-accelerated t-sne and its applications to modern data. 30th International symposium on computer architecture and high performance computing (SBAC-PAD). pp 330–338. https://doi.org/10.1109/CAHPC.2018.8645912
117. Arora S, Hu W, Kothari PK (2018) An analysis of the t-sne algorithm for data visualization. Conference on learning theory. pp 1455–1462 (**PMLR**)
118. Van Der Maaten L (2014) Accelerating t-sne using tree-based algorithms. J Mach Learn Res 15(1):3221–3245
119. De Bodt C, Mulders D, Verleysen M, Lee JA (2018) Extensive assessment of barnes-hut t-sne. ESANN
120. Demidova L, Gorchakov A (2022) Fuzzy information discrimination measures and their application to low dimensional embedding construction in the umap algorithm. J Imaging. https://doi.org/10.3390/jimaging8040113

121. Schmitz S, Weidner U, Hammer H, Thiele A (2021) Evaluating uniform manifold approximation and projection for dimension reduction and visualization of polinsar features. Remote Sensing Spatial Information Sciences, ISPRS Annals of the Photogrammetry. https://doi.org/10.5194/isprs-annals-v-1-2021-39-2021
122. Liao YT, Luo H, Ma A (2023) Efficient and robust bayesian selection of hyperparameters in dimension reduction for visualization. ArXiv **abs/2306.00357** https://doi.org/10.48550/arXiv.2306.00357
123. Kvinge H, Blumstein M (2018)Letting symmetry guide visualization: multidimensional scaling on groups. ArXiv **abs/1812.03362**
124. Lambert P, Bodt C, Verleysen M, Lee J (2022) Squadmds: a lean stochastic quartet mds improving global structure preservation in neighbor embedding like t-sne and umap. ArXiv **abs/2202.12087** https://doi.org/10.1016/j.neucom.2022.06.108
125. Charalampidis D (2023) Visualizing population structures by multidimensional scaling of smoothed pca-transformed data. IEEE Access 11:13594–13604. https://doi.org/10.1109/ACCESS.2023.3243573
126. Van Der Maaten L (2009) Learning a parametric embedding by preserving local structure. Artificial Intelligence and Statistics 384–391 (**PMLR**)
127. Gao X, Wang X (2012) Dimensionality reduction with latent variable model. Frontiers Electr Electron Eng 7:116–126. https://doi.org/10.1007/S11460-012-0179-X
128. Priam R, Nadif M (2009)A parameterization via random factor for generative topographic mapping. working paper or preprint. https://hal.science/hal-02927200
129. Bishop CM, Svensén M, Williams CKI (1998) Gtm: the generative topographic mapping. Neural Comput 10:215–234. https://doi.org/10.1162/089976698300017953
130. Liu Y, Liu Z, Li S, Yu Z, Guo Y, Liu Q, Wang G (2023) Cloud-vae: variational autoencoder with concepts embedded. Pattern Recognit 140:109530. https://doi.org/10.1016/j.patcog.2023.109530
131. Kingma DP, Welling M (2022)Auto-encoding variational Bayes. https://arxiv.org/abs/1312.6114
132. Cincilla G, Thormann M, Pons M (2010) Structuring chemical space: similarity-based characterization of the pubchem database. Mol Inf 29(1–2):37–49
133. Luque Ruiz I, Cerruela García G, Gómez-Nieto MÁ (2005) Clustering chemical databases using adaptable projection cells and mcs similarity values. J Chem Inf Model 45(5):1178–1194
134. Saeed F, Salim N, Abdo A, Hentabli H (2012) Combining multiple individual clusterings of chemical structures using cluster-based similarity partitioning algorithm. International conference on advanced machine learning technologies and applications. Springer, pp 276–284
135. Basu S, Banerjee A, Mooney RJ (2004) Active semi-supervision for pairwise constrained clustering. Proceedings of the 2004 SIAM international conference on data mining. SIAM, pp 333–344
136. MacCuish JD, MacCuish NE (2014) Chemoinformatics applications of cluster analysis. Wiley Interdisciplinary Rev Comput Mol Sci 4(1):34–48
137. Tran X-TD, Phan T-L, To V-T, Tran N-TN, Nguyen N-NS, Nguyen D-NH, Tran N-TN, Truong T (2024) Integration of the butina algorithm and ensemble learning strategies for the advancement of a pharmacophore ligand-based model: an in silico investigation of apelin agonists. Front Chem 12:1382319. https://doi.org/10.3389/fchem.2024.1382319
138. Vanitha DA, Devaraj C, Venkatesulu D, M (2014) Real coded genetic algorithm for development of optimal gk clustering algorithm. International conference on swarm, evolutionary, and memetic computing. Springer, pp 264–274
139. Mucherino A, Papajorgji PM, Pardalos PM (2009) Clustering by k-means. Data mining in agriculture. Springer, New York, pp 47–82
140. Sureja N, Chawda B, Vasant A (2022) An improved k-medoids clustering approach based on the crow search algorithm. J Comput Math Data Sci 3:100034. https://doi.org/10.1016/j.jcmds.2022.100034
141. Arora P, Deepali Varshney S (2016) Analysis of k-means and k-medoids algorithm for big data. Procedia Computer Science.78, 507–512 https://doi.org/10.1016/j.procs.2016.02.095 . 1st International Conference on Information Security & Privacy 2015
142. Digles D, Ecker GF (2011) Self-organizing maps for in silico screening and data visualization. Mol Inf 30(10):838–846. https://doi.org/10.1002/minf.201100082
143. Zhang T, Ramakrishnan R, Livny M (1996) Birch: an efficient data clustering method for very large databases. ACM SIGMOD Rec 25(2):103–114
144. Butina D (1999) Unsupervised data base clustering based on daylight's fingerprint and tanimoto similarity: a fast and automated way to cluster small and large data sets. J Chem Inf Comput Sci 39(4):747–750. https://doi.org/10.1021/ci9803381
145. Willett P (2011)Similarity searching using 2D structural fingerprints, pp. 133–158. Humana Press, Totowa, NJ. https://doi.org/10.1007/978-1-60761-839-3_5
146. MacCuish JD, MacCuish NE (2010) Clustering in bioinformatics and drug discovery. CRC Press, Boca Raton
147. Bemis GW, Murcko MA (1996) The properties of known drugs. 1. molecular frameworks. J Med Chem 39(15):2887–2893. https://doi.org/10.1021/jm9602928
148. Li Y, Hu J, Wang Y, Zhou J, Zhang L, Liu Z (2020) Deepscaffold: a comprehensive tool for scaffold-based de novo drug discovery using deep learning. J Chem Inf Model 60(1):77–91. https://doi.org/10.1021/acs.jcim.9b00727. (**PMID: 31809029**)
149. Dimova D, Stumpfe D, Hu Y, Bajorath J (2016) Analog series-based scaffolds: computational design and exploration of a new type of molecular scaffolds for medicinal chemistry. Future Sci OA 2(4):149. https://doi.org/10.4155/fsoa-2016-0058. (**PMID: 28116132**)
150. Wetzel S, Klein K, Renner S, Rauh D, Oprea TI, Mutzel P, Waldmann H (2009) Interactive exploration of chemical space with scaffold hunter. Nat Chem Biol 5(8):581–583
151. Naveja JJ, Medina-Franco JL (2019) Finding constellations in chemical space through core analysis. Front Chem. https://doi.org/10.3389/fchem.2019.00510
152. Manton KG, Lowrimore G, Yashin A, Kovtun M (2014) Fuzzy cluster analysis. Statistics Reference Online, Wiley StatsRef
153. Xue M, Zhou L, Kojima N, Muchangos LS, Machimura T, Tokai A (2018) Application of fuzzy c-means clustering to prtr chemicals uncovering their release and toxicity characteristics. Sci Total Environ 622–623:861–868. https://doi.org/10.1016/j.scitotenv.2017.12.032
154. Bezdek JC (1981)Pattern recognition with fuzzy objective function algorithms. In: Advanced Applications in Pattern Recognition. https://api.semanticscholar.org/CorpusID:30806637
155. Zhong S (2003) Probabilistic model-based clustering of complex data. The University of Texas at Austin,(PhD thesis)
156. Pikalyova R, Zabolotna Y, Horvath D, Marcou G, Varnek A(2023) The chemical library space and its application to DNA-Encoded Libraries. https://doi.org/10.26434/chemrxiv-2023-w7x9t
157. Oti EU, Olusola MO (2024) Comparative evaluation of six agglomerative hierarchical clustering methods with a robust example. Afr J Math Stat Stud 7(2):1–25
158. Roux M (2018) A comparative study of divisive and agglomerative hierarchical clustering algorithms. J Classif 35:345–366. https://doi.org/10.1007/s00357-018-9259-9
159. Kim S-S, Kwon S, Cook D (2000) Interactive visualization of hierarchical clusters using mds and mst. Metrika 51:39–51. https://doi.org/10.1007/S001840000043
160. Li N, Lefebvre N, Lengell'e R (2014) Kernel hierarchical agglomerative clustering-comparison of different gap statistics to estimate the number of clusters. International Conference on Pattern Recognition Applications and Methods, vol 2. SCITEPRESS, pp 255–262
161. Do D, Do L, McKinley SA, Terhorst J, Nguyen X (2024)Dendrogram of mixing measures: Hierarchical clustering and model selection for finite mixture models. https://arxiv.org/abs/2403.01684
162. Ferrarotti M, Decherchi S, Rocchia W (2017) Distributed kernel k-means for large scale clustering. ArXiv **abs/1710.03013** https://doi.org/10.5121/csit.2017.71015
163. Chen X (2020) K-mean clustering. Int J Comput Appl 177:24–27. https://doi.org/10.5120/ijca2020919937
164. Ahmed M, Seraj R, Islam SMS (2020) The k-means algorithm: a comprehensive survey and performance evaluation. Electronics. https://doi.org/10.3390/electronics9081295
165. Ikotun AM, Absalom EE, Abualigah L, Abuhaija B, Jia H (2022) K-means clustering algorithms: a comprehensive review, variants analysis, and advances in the era of big data. Inf Sci 622:178–210. https://doi.org/10.1016/j.ins.2022.11.139
166. Garg T, Malik A, Scholar M (2014) Survey on various enhanced k-means algorithms. International Journal of Advanced Research in Computer

and Communication Engineering. https://doi.org/10.17148/IJARCCE.2014.31138
167. Kushwaha M, Yadav H, Agrawal C (2020) A review on enhancement to standard k-means clustering. Soc Netw Comput Intell. https://doi.org/10.1007/978-981-15-2071-6_26
168. Anzali S, Gasteiger J, Holzgrabe U, Polanski J, Sadowski J, Teckentrup A, Wagener M (1998) The use of self-organizing neural networks in drug design. Perspect Drug Discovery Des 9:273–299. https://doi.org/10.1023/A:1027276425268
169. Li X, Gasteiger J, Zupan J (1993) On the topology distortion in self-organizing feature maps. Biol Cybern 70(2):189–198. https://doi.org/10.1007/BF00200832
170. Heidari J, Daneshpour N, Zangeneh A (2024) A novel k-means and k-medoids algorithms for clustering non-spherical-shape clusters non-sensitive to outliers. Pattern Recognit 155:110639. https://doi.org/10.1016/j.patcog.2024.110639
171. Huang X, Ren M, Hu Z (2020) An improvement of k-medoids clustering algorithm based on fixed point iteration. Int J Data Warehous Min 16:84–94. https://doi.org/10.4018/ijdwm.2020100105
172. Pardeshi B, Toshniwal D (2010) Improved k-medoids clustering based on cluster validity index and object density. IEEE 2nd international advance computing conference (IACC). pp 379–384. https://doi.org/10.1109/IADCC.2010.5422924
173. Schubert E, Rousseeuw PJ (2019) Faster k-medoids clustering: improving the pam, clara, and clarans algorithms. In: Amato G, Gennaro C, Oria V, Radovanović M (eds) Similarity search and applications. Springer, Cham, pp 171–187
174. Plante J (2012) Using birch to compute approximate rank statistics on massive datasets I. charest. Les Cahiers du GERAD ISSN 711:2440
175. Lorbeer B, Kosareva A, Deva B, Softic D, Ruppel P, Küpper A (2017) Variations on the clustering algorithm birch. Big Data Res 11:44–53. https://doi.org/10.1016/j.bdr.2017.09.002
176. Jiawei L (2009) An improved birch clustering algorithm and its applications. J Zhanjiang Normal College
177. Stoyanov B, Kostadinov P, Kolev M, Mustafa Z, Moskovkina M, Milina R, Bangov I (2014) Use of the descriptor fingerprints to clustering of chemical datasets. Bulg Chem Commun 46:209
178. Hernández-Hernández S, Ballester PJ (2023) On the best way to cluster nci-60 molecules. Biomolecules 13(3):498
179. Petrov KP, Bender A (2024) An open-source implementation of the scaffold identification and naming system (scins) and example applications. J Chem Inf Model 64(20):7905–7916. https://doi.org/10.1021/acs.jcim.4c01314. (**PMID: 39404472**)
180. Nasibov E, Atilgan C (2016) A note on fuzzy joint points clustering methods for large datasets. IEEE Trans Fuzzy Syst 24:1648–1653. https://doi.org/10.1109/TFUZZ.2016.2551280
181. Hadjahmadi A, Homayounpour M, Ahadi S (2012) Bilateral weighted fuzzy c-means clustering. Iranian J Electric Electron Eng 8:108–121
182. Kolen JF, Hutcheson T (2002) Reducing the time complexity of the fuzzy c-means algorithm. Trans Fuz Sys 10(2):263–267. https://doi.org/10.1109/91.995126
183. Jennath HS, Asharaf S (2022) An efficient cluster assignment algorithm for scaling support vector clustering. In: Khanna A, Gupta D, Bhattacharyya S, Hassanien AE, Anand S, Jaiswal A (eds) International conference on innovative computing and communications. Springer, Singapore, pp 285–297
184. Rathore P, Ghafoori Z, Bezdek J, Palaniswami M, Leckie C (2018) Estimating generalized dunn's cluster validity indices for big data. IEEE international conference on systems, man, and cybernetics (SMC). pp 656–661. https://doi.org/10.1109/SMC.2018.00120
185. Krishnamoorthy, R., Sreedhar Kumar, S.: An improved agglomerative clustering algorithm for outlier detection. Applied Mathematics & Information Sciences 10, 1141–1154 (2016) https://doi.org/10.18576/AMIS/100332
186. Torgo L, Soares C (2010) Resource-bounded outlier detection using clustering methods. Data Mining for Business Applications. IOS Press, Amsterdam, pp 84–98
187. Firdaus S, Uddin MA (2015) A survey on clustering algorithms and complexity analysis. Int J Comput Sci Issues (IJCSI) 12(2):62
188. Tanimoto TT (1958) An elementary mathematical theory of classification and prediction
189. Cone MM, Venkataraghavan R, McLafferty FW (1977) Computer-aided interpretation of mass spectra. 20. molecular structure comparison program for the identification of maximal common substructures. J Am Chem Soc 99(23):7668–7671. https://doi.org/10.1021/ja00465a041
190. tmap LSHForest API documentation. tmap documentation site. Accessed: 2025-12-31 (2025). https://tmap.gdb.tools/index.html
191. umap-learn (Read the Docs): Basic UMAP Parameters—Umap-learn Documentation. (2025). umap-learn (Read the Docs). Accessed: 2025-12-31. https://umap-learn.readthedocs.io/en/latest/parameters.html
192. scikit-learn: sklearn.manifold.TSNE—Scikit-learn Documentation, 1.8.0 edn. (2025). scikit-learn. Accessed: 2025-12-31. https://scikit-learn.org/stable/modules/generated/sklearn.manifold.TSNE.html
193. scikit-learn Developers: KMeans—scikit-learn 1.8.0 documentation. https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html. Accessed: 2026-01-01 (2026)
194. Wittek P, Gao SC, Lim IS, Zhao L (2017) Somoclu: an efficient parallel library for self-organizing maps. J Stat Softw 78(9):1–21. https://doi.org/10.18637/jss.v078.i09

## Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.